

Master's Degree in
**Data Science
and Advanced Analytics**

Specialization in
Data Science

**The Green Premium in the
UK Housing Market**

Evidence from Open Government Data and Machine
Learning

João Luís Pereira da Silva Marques

NOVA Information Management School
Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

**The Green Premium in the UK Housing Market: Evidence from Open
Government Data and Machine Learning**

by

João Luís Pereira da Silva Marques

Master Thesis presented as partial requirement for obtaining the Master's degree in
Data Science and Advanced Analytics, with a specialization in Data Science

Supervised by

Prof. Doutora Fátima Trindade Neves, NOVA Information Management School

April, 2026

STATEMENT OF INTEGRITY

I hereby declare having conducted this academic work with integrity. I confirm that I have not used plagiarism, any form of undue use of information or falsification of results along the process leading to its elaboration. I further declare that I have fully acknowledged the Rules of Conduct and Code of Honor from the NOVA Information Management School.

Lisbon, 28 April 2026

João Luís Pereira da Silva Marques

DEDICATION

To my parents, João Carlos and Margarida, and my sister, Rita, for being my unshakable foundation and greatest supporters.

To Francisca Salgado, for her constant presence and encouragement.

To my goddaughter, Constança, with all my love.

Dedicated also in loving memory to my grandparents. A special tribute to my grandfather, Avô Luís, whose profound emotional presence continues to guide me; to my beloved grandmothers, Avó Rosário and Avó Milú, for their endless warmth; and to my paternal grandfather, Avô Orlando, whose memory I honor and carry with me, even though I was too young to remember the moments we shared.

ACKNOWLEDGEMENTS

First and foremost, I would like to express my deepest gratitude to my advisor, Professora Doutora Fátima Trindade Neves. This dissertation would not have been possible without her unwavering support, exceptional guidance, and continuous encouragement throughout this process. Her expertise and meticulous attention to detail were fundamental in shaping the final outcome of this research.

I am also deeply grateful to Professor Doutor Miguel Castro Neto for introducing me to the Urban Analytics Lab at NOVA IMS prior to my Master's journey. His vision and the innovative work developed at the Lab were a profound source of inspiration that motivated me to enroll in this program and embark on this academic path.

I also wish to extend my sincere appreciation to Dr. Olavo Vicente and Eng. João Ferreira Lima, whose willingness to help, insightful discussions, and support during the early stages of my research journey were highly valuable and deeply appreciated.

A special acknowledgement goes to the institutions that make Open Government Data accessible, namely the HM Land Registry, the Department for Levelling Up, Housing and Communities, and the Office for National Statistics. The availability of transparent public data is the cornerstone of the empirical work developed in this thesis.

Finally, my most profound thanks go to my family and friends. Thank you for your endless patience, understanding, and for being my safe harbor during the most challenging and demanding phases of this Master's degree.

ABSTRACT

The UK's residential property market stands at the intersection of two urgent imperatives: a deepening affordability crisis and the nation's legally binding Net Zero 2050 decarbonisation agenda. However, the financial mechanisms through which energy efficiency is capitalised into property values remain imprecisely quantified, owing principally to the reliance of existing research on traditional Hedonic Pricing Models (HPMs), whose linear, fixed-coefficient architecture is constitutionally incapable of recovering the complex, non-linear spatial and structural interactions governing the UK housing market. This dissertation addresses this gap through a rigorous multi-algorithm Machine Learning pipeline, comprising Random Forest, LightGBM, and eXtreme Gradient Boosting (XGBoost), applied to a multi-source analytical dataset of 514,345 residential transactions from Greater London, Greater Manchester, and Cornwall (2021–2025), constructed through the deterministic integration of HM Land Registry Price Paid Data, Domestic Energy Performance Certificates, and ONS geospatial coordinates. XGBoost emerged as the definitive predictive champion, achieving an R^2 of 0.8249 and a Mean Absolute Error of £60,868, reducing the average prediction error by £18,581 per transaction relative to the OLS baseline ($R^2 = 0.7458$, MAE = £79,449). The SHAP (SHapley Additive exPlanations) framework was subsequently applied to the champion model to isolate the exact monetary contribution of energy efficiency to individual property valuations. The decomposition confirms the coexistence of a statistically robust "Green Premium" (+£13,506 for Band A) and a significant "Brown Discount" (−£12,017 for Band G), whilst revealing two theoretically novel findings. First, a "Historical Forgiveness" effect is identified: in specific rural heritage contexts, exemplified by a pre-1900 Cornish detached property yielding positive EPC contributions exceeding +£41,000. Second, pronounced spatial heterogeneity is documented: Greater Manchester commands the highest absolute Band A premium (+£16,818), whilst Greater London, despite imposing the nation's most severe Band G penalty (−£16,454), exhibits a compressed Green Premium consistent with a location price squeeze mechanism. These findings collectively demonstrate that blanket national energy retrofitting subsidies are geographically misallocated by design, and that effective decarbonisation policy demands regionally calibrated, property-level financial instruments of precisely the kind that the XGBoost–SHAP pipeline developed here makes possible.

KEYWORDS

Machine Learning; Explainable Artificial Intelligence (XAI); SHAP (SHapley Additive exPlanations); Open Government Data; Automated Valuation Models (AVM); Green Premium

Sustainable Development Goals (SDG):



TABLE OF CONTENTS

Statement of Integrity	ii
Dedication.....	iii
Acknowledgements	iv
Abstract	v
List of Figures	ix
List of Tables	xi
List of Abbreviations and Acronyms	xii
1. Introduction.....	1
2. Literature review	4
2.1 The Role of AI and ML in Urban Governance and Open Data	4
2.2 Traditional Statistical Methods in Property Valuation	4
2.3 Advanced Machine Learning Techniques for Automated Valuation Models.....	5
2.4 Challenges of Black-Box Models and the Need for Explainable AI	5
2.5 Identified Research Gaps and Positioning of the Current Study.....	5
3. Methodology	7
3.1 Overview of the Research Design	7
3.2 Data.....	9
3.2.1 Study Area and Geographic Delimitation	9
3.2.2 Data Sources and Temporal Scope	9
3.2.3 Dataset Construction and Variables.....	10
3.2.4 Exploratory Data Analysis (EDA) & Summary Statistics.....	11
3.2.5 Data Preprocessing & Feature Engineering	17
3.2.6 Canonical Feature Selection and Cross-Model Parity Enforcement.....	18
3.2.7 Dataset Splitting	19
3.3 Baseline model: The Hedonic Price Approach (OLS).....	19
3.4 Advanced Predictive Modelling	20
3.5 Explainability and Evaluation Metrics	22
3.5.1 Model Interpretability via SHAP.....	22
3.6 Ethical Considerations and Data Privacy	23
4. Empirical Study.....	24
4.1 Baseline Establishment (The OLS Performance)	24
4.2 Machine Learning Execution and Hyperparameter Tuning	25
4.3 Comparative Model Evaluation.....	25
5. Results and Discussion	28

5.1 Opening the Black-Box: SHAP Global Interpretability	28
5.2 The “Green Premium”: Isolating the Impact of Energy Efficiency	30
5.2.1 Intersectional Dynamics: The Historical "Forgiveness" Effect	32
5.2.2 Local Interpretability: Deconstructing Extreme Outliers.....	33
5.3 Spatial and Structural Dynamics	41
5.3.1 The Geographic Effect on the Green Premium	41
5.4 Discussion and Policy Implications.....	45
5.4.1 The Methodological Shift from Hedonic Price Models to Ensemble Learning	45
5.4.2 The Green Premium and Brown Discount: Quantification and Contextualisation	47
5.4.3 Spatial Heterogeneity and the Limits of Universal EPC Policy	48
6. Conclusions and Future Research	50
Bibliographical References	53
Appendix A - Supplementary Exploratory Data Analysis	56

LIST OF FIGURES

Figure 3.1 - End-to-End CRISP-ML Analytical Pipeline	8
Figure 3.2 - Distribution of Property Prices	13
Figure 3.3 - Correlation Heatmap (Continuous Variables).....	14
Figure 3.4 - Temporal Evolution of Average Property Prices (2021-2025)	15
Figure 3.5 - Spatial Distribution of Transactions by Price	15
Figure 3.6 - Proportion of Construction Age Bands by Region	16
Figure 3.7 - Distribution of EPC Ratings across Construction Age Bands	17
Figure 4.1 - Baseline Model Performance and Error Distribution.....	24
Figure 4.2 - Global Performance Metrics Across Models	26
Figure 4.3 - Actual vs. Predicted Property Prices (XGBoost).....	27
Figure 5.1 - Global Feature Importance (SHAP Summary Plot for the XGBoost Model)	29
Figure 5.2 - SHAP Dependence Plot: Marginal Impact of the Energy Certificate (£)	32
Figure 5.3 - SHAP Interaction Plot: Green Premium vs. Historical Age	33
Figure 5.4 - Extreme 'Brown Discount' for Property #296154	35
Figure 5.5 - Waterfall Plot: Historical Forgiveness Anomaly, Property #31531	36
Figure 5.6 - Extreme "Green Premium" (Rank 1) for Property #402155	38
Figure 5.7 - Extreme "Green Premium" (Rank 2) for Property #301818	39
Figure 5.8 - Extreme "Green Premium" (Rank 3) for Property #202067	41
Figure 5.9 - Regional Comparison - EPC Impact on Price by Region (£)	42
Figure 5.10 - Regional SHAP Summary Plots a) Greater London; b) Cornwall; c) Greater Manchester.....	44
Figure A.1 - Average property prices across Energy Performance Certificate (EPC) ratings. The non-linear trend highlights the premium on both highly efficient new builds (Bands A) and inherently inefficient historical properties (Bands E-G)....	56
Figure A.2 - Regional distribution of Energy Performance Certificate (EPC) ratings, highlighting the structural disparities between Cornwall, Greater London, and Greater Manchester.....	56
Figure A.3 - Predictive Performance Diagnostic Plot for the optimized LightGBM model: Actual versus Predicted Property Prices (£). The visualisation employs hexagonal binning to illustrate the density of observations, demonstrating a strong linear alignment along the identity line across the test set.	57
Figure A.4 - Predictive Performance Diagnostic Plot for the optimized Random Forest model: Actual versus Predicted Property Prices (£). Utilising hexagonal binning to represent transaction density, the plot reflects the model's robust predictive	

accuracy, achieving an R^2 of 0.8187 and a Mean Absolute Error of £62,201 on the
test set..... 57

LIST OF TABLES

Table 3.1 - Summary of the predictive features, variable types, and data sources, categorised by structural, spatial, and temporal dimensions.	10
Table 3.2 - Descriptive Statistics: Continuous Variables	11
Table 3.3 - Descriptive Statistics: Categorical Variables	12
Table 4.1 - Predictive Performance of the Baseline and Machine Learning Models on the out-of-sample test set	26
Table 5.1 - Mean Impact per EPC Band in £	31
Table 5.2 - Profile of the Most Penalised Property (Index #296154)	34
Table 5.3 - Profile of the Historical Forgiveness Outlier (Index #31531).....	36
Table 5.4 - Profile of the Top 1 Premium Property (Index #402155)	37
Table 5.5 - Profile of the Top 2 Premium Property (Index #301818)	39
Table 5.6 - Profile of the Top 3 Premium Property (Index #202067)	40
Table 5.7 - Regional Comparison of Energy Certificate Impact (£).....	42

LIST OF ABBREVIATIONS AND ACRONYMS

AI	Artificial Intelligence.
AVM	Automated Valuation Models.
CRISP-ML	Cross-Industry Standard Process for Machine Learning.
DL	Deep Learning.
EDA	Exploratory Data Analysis.
EPC	Energy Performance Certificates.
GPS	Global Positioning System.
HPM	Hedonic Pricing Model.
IQR	Interquartile Range.
LightGBM	Light Gradient Boosting Machine.
LR-PPD	Land Registry Price Paid Data.
MAE	Mean Absolute Error.
MAPE	Mean Absolute Percentage Error.
ML	Machine Learning.
MLP	Multilayer Perceptron.
OD	Open Data.
OGD	Open Government Data.
OLS	Ordinary Least Squares.
ONS	Office for National Statistics.
PII	Personally Identifiable Information.
POI	Points of Interest.
RF	Random Forest.
RMSE	Root Mean Squared Error.
R²	Coefficient of Determination.
SC	Smart Cities.
SHAP	SHapley Additive exPlanations.

TPE	Tree-structured Parzen Estimator.
UK	United Kingdom.
XAI	eXplainable Artificial Intelligence.
XGBoost	eXtreme Gradient Boosting.



1. INTRODUCTION

Accurate residential property valuation is a fundamental component of urban planning, real estate investment, and effective policymaking (Allam & Dhunny, 2019; Bibri et al., 2023; Deng & Zhang, 2025; Hong & Kim, 2022). As the real estate market grows increasingly complex, stakeholders such as appraisers, mortgage lenders, and policymakers demand Automated Valuation Models (AVMs) that can process large, dynamic datasets with high precision (Hong & Kim, 2022; Tchuente, 2024; Tekouabou et al., 2024; Zhao et al., 2024). Historically, the industry has relied on traditional econometric methods, most notably the Hedonic Pricing Model (HPM) (Geerts et al., 2023; Hong & Kim, 2022; Zhao et al., 2024). HPMs evaluate property prices as a linear combination of structural, neighbourhood, and locational attributes, offering high native interpretability through beta coefficients and p-values (Deng & Zhang, 2025; Fik et al., 2003; Geerts et al., 2023; Tarasov & Dessoulavy-Śliwiński, 2025; Tchuente, 2024). However, these traditional multiple regression analyses impose severe theoretical constraints, such as strict assumptions of linearity, homoscedasticity, and the absence of multicollinearity, which limit their ability to capture the complex, non-linear realities of dynamic housing markets (Deng & Zhang, 2025; Feng et al., 2021; Geerts et al., 2023; Hong & Kim, 2022; Tarasov & Dessoulavy-Śliwiński, 2025). This question carries particular urgency given the UK Government's legally binding commitment to achieving Net Zero carbon emissions by 2050, which requires substantial improvements to the energy efficiency of the existing residential housing stock.

To overcome the limitations of traditional econometrics, there has been a significant methodological shift toward Machine Learning (ML) and Artificial Intelligence (AI) in real estate analytics (Deng & Zhang, 2025; Geerts et al., 2023; Hong & Kim, 2022; Tekouabou et al., 2024). Advanced ML algorithms, particularly tree-based ensemble methods like Random Forest (RF), LightGBM, and eXtreme Gradient Boosting (XGBoost), as well as Deep Learning (DL) architectures like Multilayer Perceptrons (MLP), have consistently demonstrated superior out-of-sample predictive accuracy (Deng & Zhang, 2025; Fonseca et al., 2023; Geerts et al., 2023; Hong & Kim, 2022; Tarasov & Dessoulavy-Śliwiński, 2025; Tchuente, 2024). These data-driven models excel at identifying complex, multilevel interactions and non-linear associations without requiring pre-specified functional forms (Hong & Kim, 2022; Taecharungroj, 2021). However, their widespread adoption by industry practitioners has been hindered by their "black-box" nature (N. Chen, 2022; Oust et al., 2023; Tarasov & Dessoulavy-Śliwiński, 2025; Tchuente, 2024; Tekouabou et al., 2024). To foster trust and ensure accountability, there is a critical need to integrate eXplainable Artificial Intelligence (XAI) frameworks, such as SHapley Additive exPlanations (SHAP), which can decode complex ML decisions and quantify the precise marginal contribution of individual features to property values (Deng & Zhang, 2025; Karamanou et al., 2022; Tarasov & Dessoulavy-Śliwiński, 2025; Tchuente, 2024; Trindade Neves et al., 2024).



The success of these advanced predictive models is inherently tied to the quality and granularity of the underlying data (Bibri et al., 2023; Kalampokis et al., 2018; Karamanou et al., 2022; Tekouabou et al., 2024; Trindade Neves et al., 2024; Zhao et al., 2024). While previous studies heavily relied on proprietary commercial data, the modern paradigm emphasises the immense value of Open Urban Data (Bibri et al., 2023; Kang et al., 2021; Karamanou et al., 2022; Trindade Neves et al., 2024; Zhao et al., 2024). In the context of the United Kingdom, the integration of Open Government Data (OGD), specifically the UK Land Registry Price Paid Data (LR-PPD) and Domestic Energy Performance Certificates (EPC), has revolutionised housing market research (Chi et al., 2021). By combining comprehensive transaction records with precise structural attributes and ONS (Office for National Statistics) geospatial data, researchers can build highly nuanced models (Chi et al., 2021). This multi-source data fusion is especially vital for capturing emerging market dynamics, such as the "green premium," where buyers increasingly capitalise on energy efficiency (EPC ratings) amidst growing environmental awareness and rising energy costs (Chi et al., 2021; Fonseca et al., 2023; Wenninger et al., 2022).

Despite the proven advantages of ML/AI and Open Urban Data, significant research gaps remain in systematically benchmarking these advanced models against traditional spatial econometrics across diverse geographical landscapes, as well as in explicitly extracting non-linear environmental premiums. To address these gaps, this thesis pivots to the UK housing market to explore the intersection of predictive accuracy, model transparency, and spatial robustness.

Accordingly, the core objectives of this research are explicitly structured as follows:

- **Objective 1: Performance Comparison:** Compare the out-of-sample predictive performance (measured via MAE, RMSE, R^2 and MAPE) of traditional Hedonic Models (OLS Regression) against advanced tree-based ML models (Random Forest, LightGBM, XGBoost). Both approaches will utilise One-Hot Encoding for categorical variables. To capture spatial dependencies, the Hedonic models will employ postcode Outcodes as Spatial Fixed Effects, whereas the ML models will leverage exact latitude and longitude coordinates to automatically learn complex spatial interactions.
- **Objective 2: Interpretability:** Contrast the native explainability of Hedonic models (relying on traditional beta coefficients and p-values) with post-hoc SHAP values extracted from the ML models. This comparison will specifically focus on uncovering and interpreting non-linear financial premiums, particularly the "green premium" associated with EPC energy efficiency ratings, thereby bridging the gap between high predictive accuracy and stakeholder transparency.
- **Objective 3: Robustness:** Validate the generalisability and robustness of the developed models across three distinctly different spatial markets: Greater Manchester (serving as the urban baseline), Greater London (acting as a hyper-dense stress test to evaluate model performance amidst extreme outliers and



premium pricing), and Cornwall (a rural/coastal environment to test the efficacy of spatial proxies in areas with a scarcity of Points of Interest).

This dissertation is structured as follows. Chapter 2 reviews the relevant literature on traditional hedonic pricing models, advanced machine learning approaches in real estate valuation, and the role of explainable artificial intelligence in improving model transparency, culminating in the identification of the research gaps addressed in this study. Chapter 3 outlines the methodological framework, including the data sources, preprocessing procedures, feature engineering strategy, and the development of both the baseline econometric model and the machine learning pipeline, as well as the evaluation and explainability techniques employed. Chapter 4 presents the empirical study, detailing model training, hyperparameter optimisation, and the comparative performance analysis across all approaches. Chapter 5 reports and discusses the main findings, with a particular focus on the SHAP-based decomposition of property values, the quantification of the Green Premium and Brown Discount, and the analysis of spatial and structural heterogeneity. Finally, Chapter 6 concludes the dissertation by summarising the key contributions, discussing theoretical and practical implications, acknowledging limitations, and outlining directions for future research.



2. LITERATURE REVIEW

2.1 THE ROLE OF AI AND ML IN URBAN GOVERNANCE AND OPEN DATA

The growing complexity of urban governance and the development of smart cities increasingly demand the implementation of AI and ML solutions, which are central to managing urban growth, automating operations, and supporting evidence-based decision-making for policymakers (Allam & Dhunny, 2019; Neves et al., 2020; Tekouabou et al., 2024; Trindade Neves et al., 2024; Yigitcanlar et al., 2021). A vital enabler for transparent urban development and effective AI modelling is the availability of OGD, which provides crucial statistical data relating to economic, social, and environmental domains (Bibri et al., 2023; Karamanou et al., 2022; Tekouabou et al., 2024; Trindade Neves et al., 2024). In the United Kingdom, the release of comprehensive open datasets has been transformative for real estate analytics, exemplified by the official LR-PPD (Chi et al., 2021). According to the framework established by Chi et al. (2021), combining these transaction records with Domestic EPC, which offer detailed property attributes such as total floor area and energy efficiency ratings, and ONS geospatial data, allows researchers to build highly nuanced, multi-source predictive models. This linked open data paradigm facilitates the efficient integration of diverse datasets, capturing the complex urban dynamics necessary for accurate real estate valuation (Deng & Zhang, 2025; Hong & Kim, 2022; Kalampokis et al., 2018; Karamanou et al., 2022).

2.2 TRADITIONAL STATISTICAL METHODS IN PROPERTY VALUATION

Historically, residential property valuation has relied heavily on the HPM, rooted in the premise that a differentiated product is completely described by a vector of objectively measured characteristics, with observed market prices defining a set of implicit or "hedonic" prices (Geerts et al., 2023; Rosen, 1974). Typically estimated using Ordinary Least Squares (OLS) or panel regressions, the traditional hedonic approach assumes that buyers evaluate structural and locational characteristics separately, treating location as an independent premium added to the base price of the home (Fik et al., 2003). However, this traditional econometric framework imposes severe constraints, primarily the strict assumption of linearity, which fails to capture complex spatial realities, as Fik et al. (2003) emphasise that the marginal value of housing characteristics varies continuously over space. Therefore, non-interactive, additive models often misspecify the relationship by erroneously holding the price of land spatially constant across urban areas. Furthermore, traditional spatial methods struggle with spatial autocorrelation and multicollinearity (Cellmer, 2023; Fik et al., 2003). As demonstrated by Wheeler and Tiefelsdorf (2005), standard local regression frameworks designed to account for spatial heterogeneity can suffer from severe multicollinearity among local regression coefficients, which distorts the estimation of spatial dependencies and implicit prices.



2.3 ADVANCED MACHINE LEARNING TECHNIQUES FOR AUTOMATED VALUATION MODELS

To overcome the limitations of traditional econometrics, the real estate literature has shifted towards advanced ML techniques, which can effectively capture complex, non-linear relationships without requiring pre-specified functional forms (Deng & Zhang, 2025). Algorithms such as Random Forest, eXtreme Gradient Boosting (XGBoost), LightGBM, and Artificial Neural Networks like the MLPs have become front-runners in maximising predictive accuracy (Deng & Zhang, 2025; Fonseca et al., 2023). XGBoost offers a highly scalable end-to-end tree boosting system that utilises an objective function combining model deviation and a regularization term to prevent overfitting, while efficiently handling large, complex datasets by leveraging parallel computing and sparsity-aware split finding (T. Chen & Guestrin, 2016; Deng & Zhang, 2025). Similarly, MLPs establish non-linear mappings across multiple computational layers to achieve precise property valuations (Fonseca et al., 2023). Previous studies have successfully utilised these ML models to estimate the sale prices of real estate units with exceptionally high accuracy compared to baseline linear models (Li et al., 2021). In the UK context, researchers like Chi et al. (2021) have demonstrated the immense value of integrating LR-PPD transaction data with EPC size data to generate a rich dataset for modelling house price variations. These data-driven tree ensembles consistently outperform traditional methods by automatically learning the hierarchical interactions of structural features and locational attributes (Hong & Kim, 2022).

2.4 CHALLENGES OF BLACK-BOX MODELS AND THE NEED FOR EXPLAINABLE AI

Despite their superior predictive capabilities, advanced ML algorithms are often criticised for their “black-box” nature, which obscures the internal logic behind their predictions (Tarasov & Dessoulavy-Śliwiński, 2025). This lack of interpretability poses a significant barrier to their adoption by real estate stakeholders, mortgage lenders, and urban policymakers who require transparent, factual, and verifiable explanations for property valuations (Tchunte, 2024; Tekouabou et al., 2024). To address this transparency deficit, SHAP has emerged as the preferred XAI framework, as it calculates the exact marginal contribution of each feature to a specific prediction based on cooperative game theory, enabling both local and global interpretability (Deng & Zhang, 2025; Karamanou et al., 2022; Oust et al., 2023; Tarasov & Dessoulavy-Śliwiński, 2025; Tchunte, 2024; Trindade Neves et al., 2024). Furthermore, Wenninger et al. (2022) highlight the value of combining XGBoost and SHAP to analyse the UK's residential building stock, illustrating how XAI can decipher complex AI predictions to uncover critical socio-economic relationships and explicitly quantify the financial premiums associated with energy-efficiency measures.

2.5 IDENTIFIED RESEARCH GAPS AND POSITIONING OF THE CURRENT STUDY

While existing literature confirms the efficacy of ML-based AVMs and the growing relevance of XAI, several methodological and contextual gaps remain in the study of the UK housing market. First, most studies using UK open data aggregate locational



attributes to coarse geographic boundaries, such as postcode Outcodes or census tracts, missing the hyper-local price variations that exact GPS coordinates (latitude and longitude) can capture in spatial ML models (Chi et al., 2021). Second, although the general impact of energy efficiency has been investigated (Wenninger et al., 2022), there is a distinct gap in evaluating the post-pandemic "green premium" explicitly extracted through XAI frameworks like SHAP using recently updated EPC and transaction data. Finally, many AVM studies restrict their focus to a single urban market, which severely limits the generalisability and structural validation of their models across different spatial topologies (Deng & Zhang, 2025).

This thesis is fundamentally positioned to address these gaps. It will explicitly compare the out-of-sample predictive performance of exact GPS coordinates in non-linear ML models against traditional spatial fixed effects (Outcodes) in Hedonic models. Furthermore, it will employ SHAP to systematically isolate and interpret the non-linear financial premiums associated with post-pandemic EPC ratings. Lastly, the research will rigorously test model robustness across three distinctly diverse regional settings: the urban baseline of Greater Manchester, the hyper-dense stress test of Greater London, and the rural/coastal environment of Cornwall.



3. METHODOLOGY

The complete codebase, including datasets, analytical notebooks, and implementation details, is publicly available in a GitHub repository¹ to ensure full transparency and reproducibility of the results.

3.1 OVERVIEW OF THE RESEARCH DESIGN

To ensure that the predictive modelling process is robust, transparent, and fully reproducible, this study adopts a quantitative, data-driven research design structured around the Cross-Industry Standard Process for ML (CRISP-ML) framework. The CRISP-ML methodology is an industry-standard paradigm for end-to-end ML, providing a systematic approach to transitioning from raw data acquisition to model explainability.

Accordingly, the analytical pipeline is divided into three primary phases:

Phase 1 – Data Transformation and Analysis:

Encompassing data collection from multiple open government sources, rigorous preprocessing (including the handling of missing values, outlier detection, and geospatial linkage), and comprehensive Exploratory Data Analysis (EDA).

Phase 2 – Modelling Strategy:

Establishing a performance baseline using traditional HPMs (OLS) with postcode Outcodes, followed by the training and hyperparameter optimisation of advanced, non-linear ML algorithms (Random Forest, LightGBM, and XGBoost) using exact GPS coordinates.

Phase 3 – Explainability and Evaluation:

Assessing predictive accuracy through robust error metrics (MAE, RMSE, R^2 and MAPE) and applying post-hoc XAI, specifically SHAP, to extract actionable insights and quantify the "green premium".

Figure 3.1 provides a visual representation of this end-to-end pipeline, from raw data ingestion through to the final SHAP economic outputs.

¹ <https://github.com/jm651120/UK-Housing-Green-Premium>

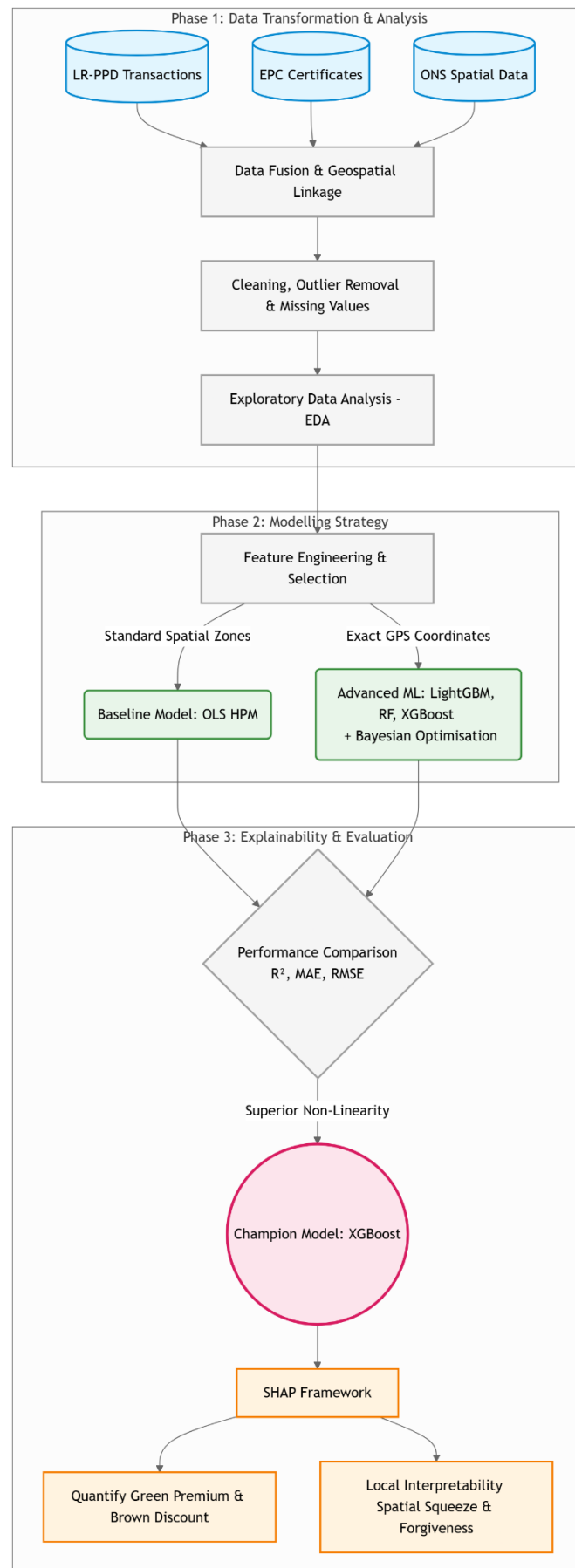


Figure 3.1 - End-to-End CRISP-ML Analytical Pipeline



3.2 DATA

3.2.1 STUDY AREA AND GEOGRAPHIC DELIMITATION

To provide a comprehensive and robust analysis of the UK housing market, this study focuses on three distinct geographic regions: Greater London, Greater Manchester, and Cornwall. This selection was strategically made to capture contrasting market dynamics. Greater London represents a highly dense, high-value global economic hub with unique pricing behaviours. Greater Manchester serves as a major, rapidly developing urban agglomeration in the North of England, offering a comparative metropolitan perspective. Conversely, Cornwall introduces a rural and coastal dimension, characterised by different housing typologies, seasonal demand, and distinct socioeconomic factors.

3.2.2 DATA SOURCES AND TEMPORAL SCOPE

The empirical foundation of this study relies entirely on OGD, ensuring both transparency and reproducibility. The dataset is a multi-source fusion constructed from three primary official providers:

- **HM Land Registry Price Paid Data (LR-PPD)²:** This official dataset provides a comprehensive, address-level record of residential property transactions in England and Wales. It serves as the primary source for the target variable (sale price) and fundamental transaction details (e.g., date of sale, property type, and tenure).
- **Domestic Energy Performance Certificates (EPC)³:** Managed by the Ministry of Housing, Communities and Local Government (MHCLG), this dataset offers highly granular intrinsic structural attributes, including total floor area, property age, heating systems, and the critical energy efficiency ratings required to assess the "green premium".
- **Office for National Statistics (ONS)⁴:** The ONS Open Geography Portal provides the essential geospatial linkages, mapping property postcodes to exact latitude and longitude coordinates, thereby enabling the extraction of hyper-local spatial variations.

To capture the most recent market dynamics and specifically address the research gap regarding the post-pandemic housing market, the temporal scope of the data collection is strictly bound to transactions occurring between January 2021 and 2025. This specific timeframe is critical, as it encompasses the post-COVID-19 market

² HM Land Registry Price Paid Data. Available at: <https://www.gov.uk/government/statistical-data-sets/price-paid-data-downloads> (Accessed: February 2026).

³ Department for Levelling Up, Housing and Communities. Energy Performance of Buildings Data. Available at: <https://epc.opendatacommunities.org/> (Accessed: February 2026). Note: This service has recently been migrated to the Ministry of Housing, Communities and Local Government (MHCLG) portal at: <https://get-energy-performance-data.communities.gov.uk/>

⁴ Office for National Statistics ONS Postcode Directory (November 2025) for the UK. Open Geography Portal Available at: <https://geoportal.statistics.gov.uk/datasets/3635ca7f69df4733af27caf86473ffa1/about> (Accessed: February 2026).



adjustments and the period of heightened environmental awareness, allowing the models to capture the modern “green premium” under contemporary macroeconomic conditions.

3.2.3 DATASET CONSTRUCTION AND VARIABLES

To construct the final analytical dataset, a deterministic data linkage strategy was employed. The transaction records (LR-PPD) and energy certificates (EPC) were joined using the property's alphanumeric postcode as the primary spatial key. To resolve duplicates, where multiple transactions or certificates exist for a single postcode, address string matching (utilising PAON/Street from LR-PPD and ADDRESS1 from EPC) and temporal proximity rules were applied, ensuring a one-to-one mapping between a sale price and its corresponding energy rating.

Following the integration, the predictive features were selected and structured into a comprehensive variable table (Table 3.1). As mandated by the research design, these features are categorised into intrinsic structural attributes, spatial/extrinsic factors, and temporal/macroeconomic controls, as detailed below:

Table 3.1 - Summary of the predictive features, variable types, and data sources, categorised by structural, spatial, and temporal dimensions.

Category	Variable	Type	Description	Source
Target Variable	Price	Continuous	Final registered transaction price of the property, in GBP (£). Serves as the dependent variable across all models.	LR-PPD
Intrinsic / Structural	Property_Type	Categorical	Architectural classification of the dwelling: Detached, Semi-Detached, Terraced, or Flat/Maisonette.	LR-PPD
Intrinsic / Structural	Old_New	Categorical	Indicates whether the property is a newly built home (Y) or an established residential dwelling (N).	LR-PPD
Intrinsic / Structural	Tenure_Duration	Categorical	Legal ownership type: Freehold (F) or Leasehold (L). A critical legal and financial distinction in the UK property market.	LR-PPD
Intrinsic / Structural	Total_Floor_Area	Continuous	Total usable internal floor space, measured in square metres. Primary structural proxy for property scale.	EPC
Intrinsic / Structural	Number_Habitable_Rooms	Continuous	Count of rooms designated for habitation. Acts as a secondary proxy for property capacity and household size.	EPC
Intrinsic / Structural	Current_Energy_Rating	Ordinal	EPC energy efficiency band (A to G), ordinal encoded as A=7 through G=1. The primary feature for quantifying the green premium.	EPC
Intrinsic / Structural	Construction_Age_Band	Categorical	Historical construction era of the dwelling (e.g., pre-1900, 1900–1929, 2007–2011). Critical confounding variable for energy performance.	EPC



Category	Variable	Type	Description	Source
Spatial / Locational	Postcode_Outcode	Categorical	First segment of the UK postcode (e.g., SW1, M14). Used exclusively in the OLS model as a spatial fixed effect to absorb sub-market price variation.	LR-PPD
Spatial / Locational	Latitude	Continuous	Exact geographic latitude coordinate. Used by ML models to learn hyper-local, non-linear spatial price boundaries.	ONS
Spatial / Locational	Longitude	Continuous	Exact geographic longitude coordinate. Used by ML models to learn hyper-local, non-linear spatial price boundaries.	ONS
Temporal / Macroeconomic	Year_of_Transfer	Categorical	Year of the transaction. Controls for annual macroeconomic shifts, including post-pandemic rate cycles (2021–2025).	LR-PPD
Temporal / Macroeconomic	Month_of_Transfer	Categorical	Calendar month of the transaction. Controls for intra-annual seasonal demand patterns.	LR-PPD

3.2.4 EXPLORATORY DATA ANALYSIS (EDA) & SUMMARY STATISTICS

Following data integration, an Exploratory Data Analysis (EDA) was conducted to investigate the underlying distributions, identify potential skewness in the target variable, and examine foundational correlations between intrinsic features and property prices. A core component of this analysis was visualising the price variations across different EPC bands to identify preliminary evidence of the "green premium". Table 3.2 presents the summary statistics for the continuous variables in the final pre-processed dataset, providing a macro-level overview of the housing market dynamics captured within the sample.

Table 3.2 - Descriptive Statistics: Continuous Variables

Variable	N	Mean	Std Dev	Min	Median	Max
Price (£)	514,345	412,935	227,361	50,000	380,000	1,162,500
Total Floor Area (m ²)	514,345	83.15	26.53	15	80	158.22
Number of Habitable Rooms	514,345	4.23	1.29	1	4	10
Latitude	514,345	52.0632	1.0462	49.9652	51.5566	53.6801
Longitude	514,345	-1.1759	1.4763	-5.705	-0.2848	0.3094
EPC Rating (encoded 1–7)	514,345	4.19	0.94	1	4	7



Table 3.3 - Descriptive Statistics: Categorical Variables

Variable	Level	N	Proportion (%)
EPC Rating	Band A	1,476	0.30%
	Band B	34,732	6.80%
	Band C	143,210	27.80%
	Band D	240,565	46.80%
	Band E	74,916	14.60%
	Band F	14,228	2.80%
	Band G	5,218	1.00%
Property Type	Detached	46,036	9.00%
	Flat / Maisonette	134,367	26.10%
	Other	12,329	2.40%
	Semi-Detached	130,069	25.30%
	Terraced	191,544	37.20%
Old / New Build	Established (N)	489,445	95.20%
	New Build (Y)	24,900	4.80%
Tenure Duration	Freehold	306,977	59.70%
	Leasehold	207,368	40.30%
Year of Transfer	2021	135,333	26.30%
	2022	114,123	22.20%
	2023	90,433	17.60%
	2024	98,917	19.20%
	2025	75,539	14.70%
Month of Transfer	Jan	37,165	7.20%
	Feb	41,332	8.00%
	Mar	58,063	11.30%
	Apr	35,700	6.90%
	May	38,094	7.40%
	Jun	55,647	10.80%
	Jul	39,623	7.70%
	Aug	42,783	8.30%
	Sep	48,107	9.40%
	Oct	40,671	7.90%
	Nov	40,122	7.80%
	Dec	37,038	7.20%
Construction Age Band	Before 1900	54,022	10.50%
	1900–1929	117,912	22.90%
	1930–1949	119,098	23.20%
	1950–1966	59,795	11.60%
	1967–1975	37,570	7.30%
	1976–1982	19,428	3.80%
	1983–1990	22,227	4.30%
	1991–1995	12,543	2.40%
	1996–2002	17,010	3.30%
	2003–2006	13,519	2.60%
	2007 onwards	41,221	8.00%



The Construction Age Band categories displayed in Table 3.3 have been harmonised into the eleven canonical age band designations employed by the Energy Performance Certificate (EPC) register for England and Wales, solely for the purpose of producing a legible and interpretable descriptive overview. A minority of source records contained numeric construction year entries (e.g., 2013–2025) and overlapping post-2007 label variants (e.g., England and Wales: 2007–2011, England and Wales: 2012 onwards) arising from heterogeneity in EPC data submissions. In the summary table, these entries have been consolidated into their nearest canonical band to avoid an 86-row distribution in which most rows carry negligible proportions. It must be emphasised, however, that this harmonisation was applied exclusively for presentational purposes and was not imposed upon the modelling pipeline.

The machine learning algorithms, LightGBM, Random Forest, and XGBoost, were trained on the raw, un-harmonised Construction Age Band labels, which were encoded directly via One-Hot Encoding with a dropped reference category (`drop_first=True`). Consequently, stray numeric year entries and overlapping post-2007 label variants were retained as distinct binary dummy variables, yielding 27 Construction Age Band features within the final 53-feature model space. This design decision was deliberate: by preserving the full granularity of the source data, the algorithms were afforded the opportunity to identify systematic price variation associated with specific label values, rather than having an a priori aggregation imposed that might suppress latent heterogeneity.

As illustrated in Figure 3.2, the distribution of property prices is highly right-skewed, demonstrating a dense concentration of transactions in the lower-to-mid price brackets, accompanied by a long tail of high-value outliers. This structural skewness justifies the subsequent application of robust outlier detection methods and highlights the non-linear challenges the predictive models will face.

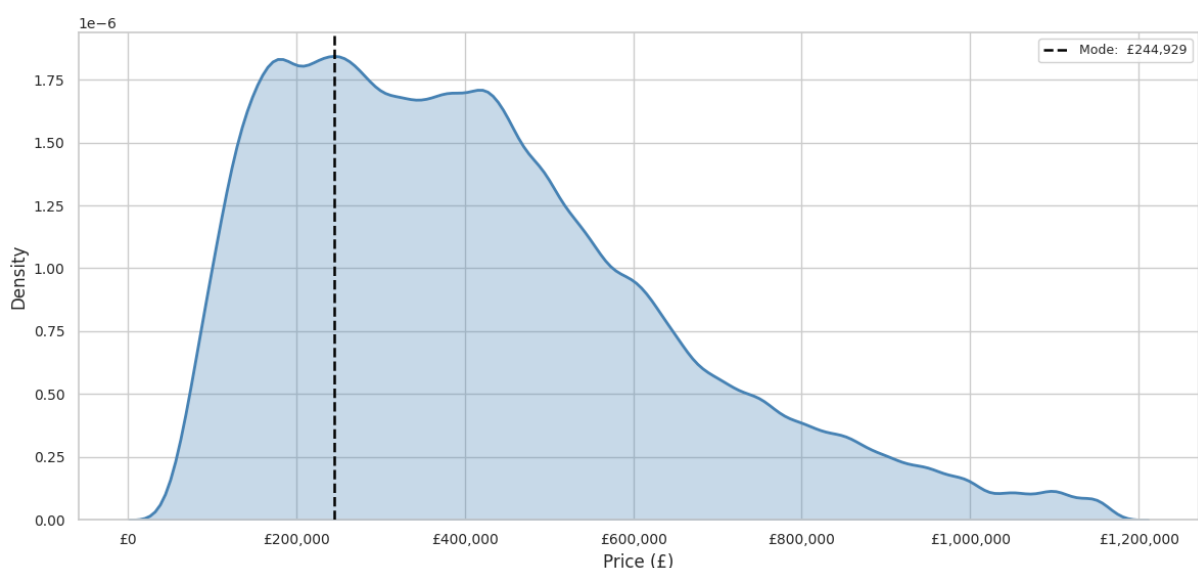


Figure 3.2 - Distribution of Property Prices



Preliminary visual analysis (detailed in Figure A.1, Appendix A) indicates that while Band A properties command the highest average prices, a non-linear relationship exists. This is likely driven by high-value historical and heritage properties populating the lower EPC bands (E-G).

Furthermore, the correlation matrix (Figure 3.3) highlights that Latitude (-0.46) and Total Floor Area (0.39) are significant drivers of value, reinforcing the need for multi-variable predictive modelling.



Figure 3.3 - Correlation Heatmap (Continuous Variables)

Building upon these preliminary findings, it is essential to contextualize the market's macroeconomic and geographical heterogeneity. The UK housing market between 2021 and 2025 experienced significant dynamic shifts due to post-pandemic recovery and inflation. Furthermore, the geographical scope of this study encompasses highly distinct markets, ranging from the dense urban core of Greater London to the rural coastal areas of Cornwall.

As illustrated in Figure 3.4, average property prices exhibit clear temporal fluctuations, justifying the inclusion of the transaction year and month as control variables in the modelling phase.

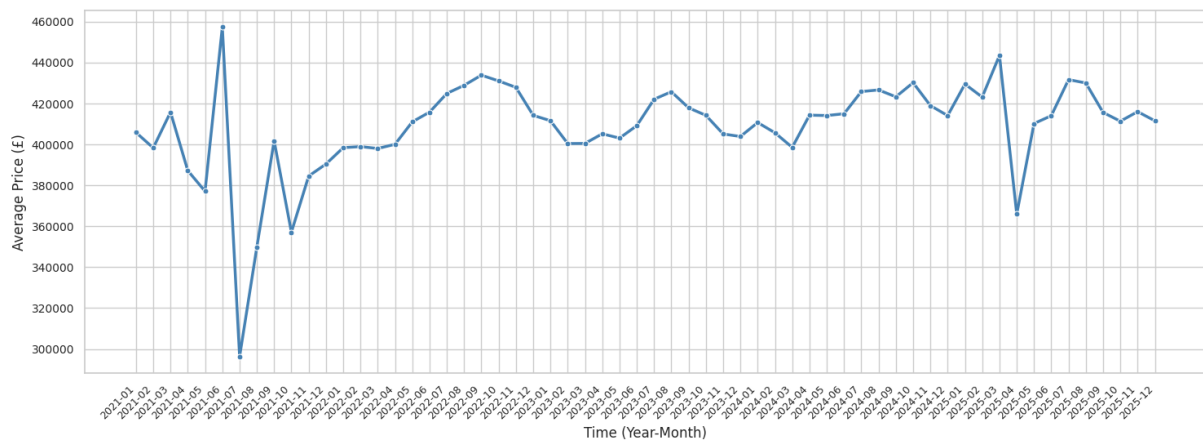


Figure 3.4 - Temporal Evolution of Average Property Prices (2021-2025)

Simultaneously, the spatial distribution (Figure 3.5) provides visual confirmation of the extreme price density in London compared to the other regions, corroborating the negative correlation found with Latitude and mandating the use of spatial fixed effects (*Outcodes*) in the Hedonic model.

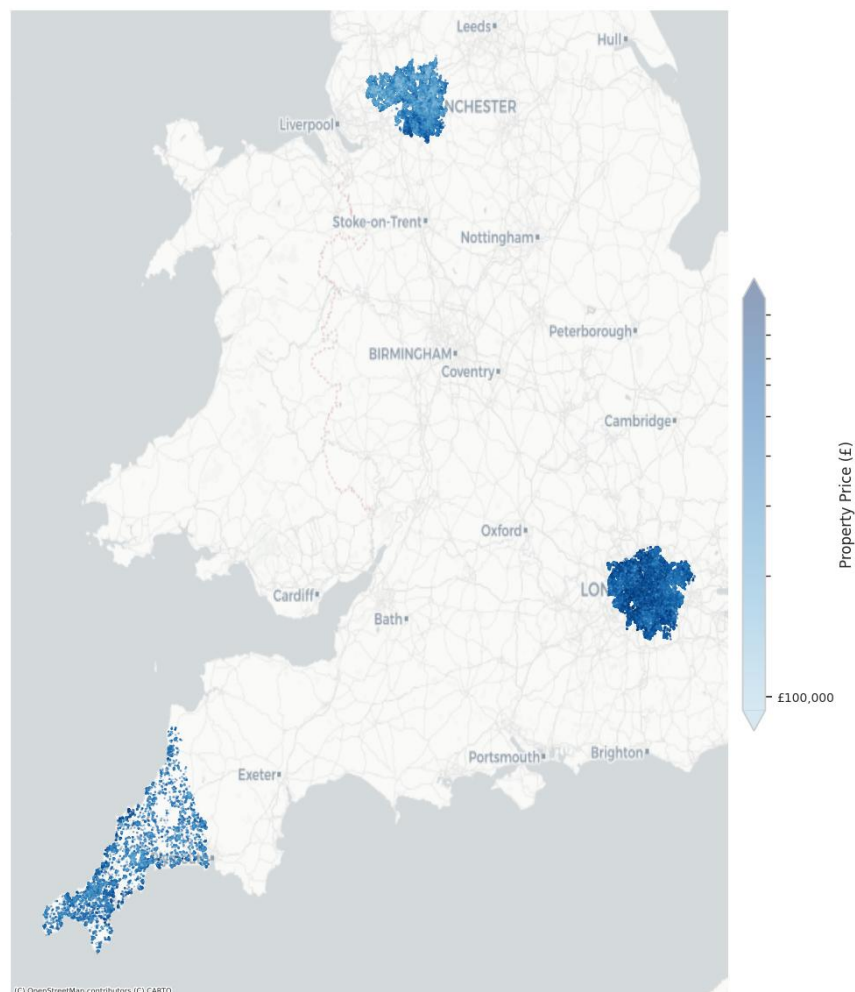


Figure 3.5 - Spatial Distribution of Transactions by Price



Finally, to empirically validate the hypothesis regarding high-value historical properties skewing the lower EPC bands, an analysis of energy efficiency profiling was conducted across regions and construction age bands.

As observed in Figure 3.6, the structural age of the housing stock varies significantly across regions. Greater London exhibits the highest proportion of older properties, with a heavy concentration built between 1900 and the 1930s, supporting the "historical premium" hypothesis. Conversely, Greater Manchester displays a more uniformly distributed and balanced age profile. Cornwall presents a highly polarised market: it holds a substantial proportion of historical properties from the 1800s but also leads in modern constructions (post-2016). (Note: The minor missing percentages in Cornwall represent records with 'Unknown' or 'Invalid' age data).

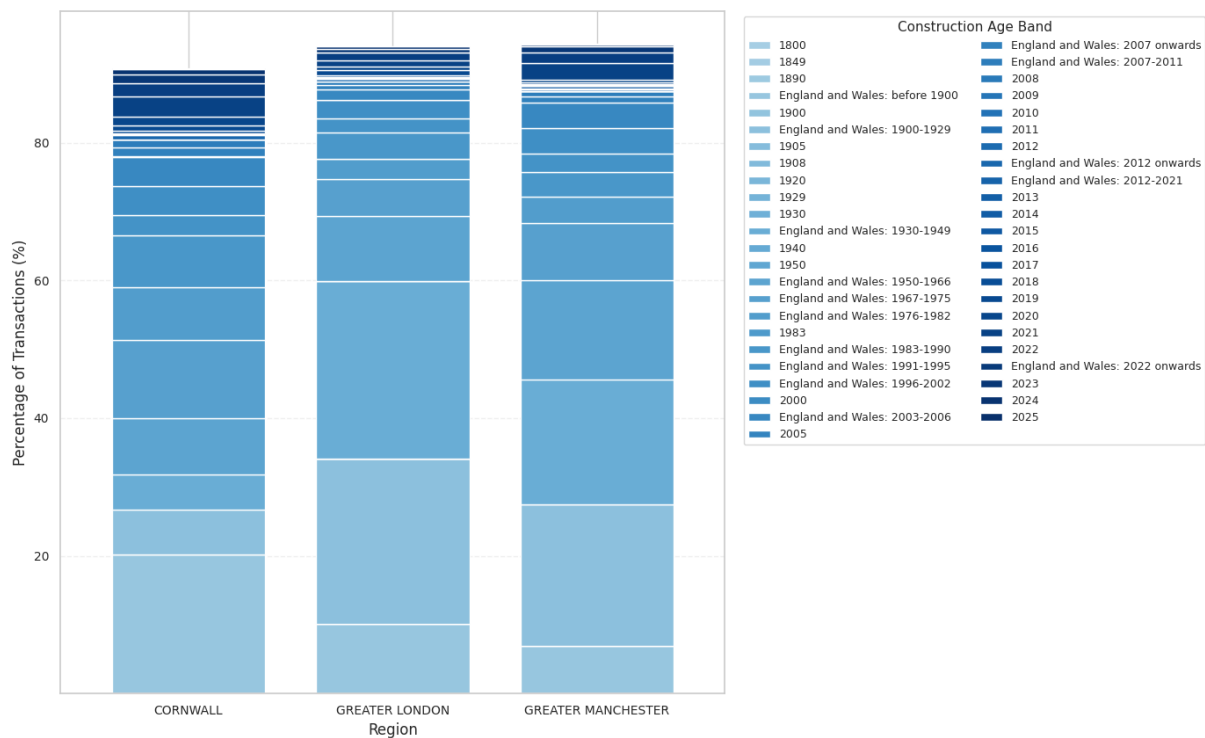


Figure 3.6 - Proportion of Construction Age Bands by Region

This structural age disparity directly impacts the regional energy profiles (as detailed in Figure A.2, Appendix A). Cornwall's polarised age distribution translates into a polarised EPC distribution: it has a higher proportion of highly efficient homes (Ratings A and B, driven by recent builds) but also a heavier tail of highly inefficient properties (Ratings E, F, and G, from the 1800s stock), with notably fewer mid-tier D ratings. Meanwhile, Greater London and Greater Manchester exhibit remarkably similar overall EPC proportions, though London shows a slight increase in Band G properties, directly aligning with its vast pre-war housing stock.



To conclusively tie energy performance to historical building regulations, Figure 3.7 maps EPC ratings against property age. The heatmap reveals a stark chronological evolution. Properties built prior to the 1980s are overwhelmingly concentrated in the 'D' band or lower. A structural improvement becomes evident between 1983 and 1990, where most properties begin achieving 'C' ratings. Finally, properties constructed from 2007 onwards show a massive shift towards 'B' ratings, reflecting modern, stricter building regulations.

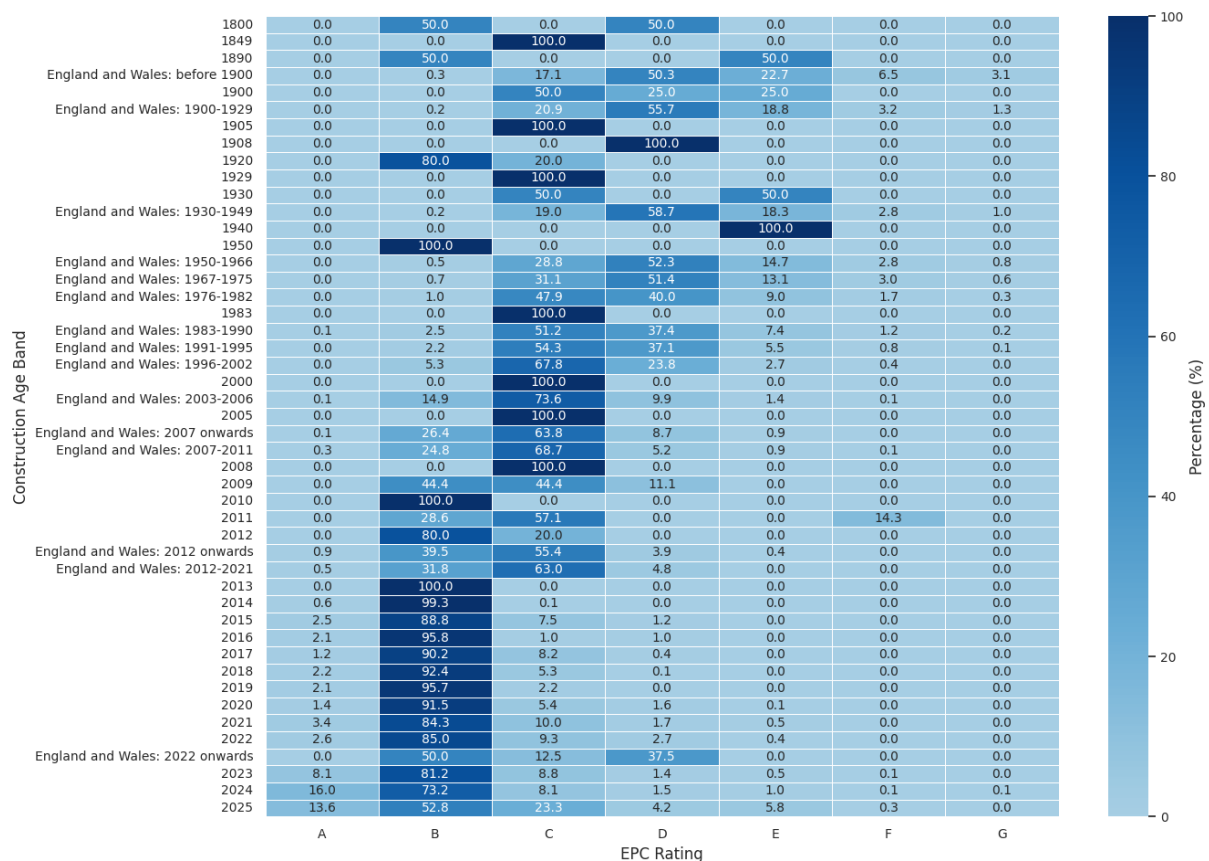


Figure 3.7 - Distribution of EPC Ratings across Construction Age Bands

Ultimately, this visual evidence confirms that *Construction_Age_Band* and geography are critical confounding variables. Strictly controlling for these factors, via spatial fixed effects and chronological features, is mandatory in the subsequent Hedonic Pricing and ML models. Only by decoupling a property's inherent historical and locational value can the true "green premium" be accurately isolated and evaluated.

3.2.5 DATA PREPROCESSING & FEATURE ENGINEERING

In alignment with the CRISP-ML framework, rigorous data preprocessing and feature engineering were conducted to ensure the robustness of the predictive models. Missing values primarily affected *Number_Habitable_Rooms* (70,739 instances) and *Construction_Age_Band* (6,398 instances). To maintain distributional integrity, missing continuous values were imputed using the sample median, while categorical gaps were



reclassified as "Unknown". Minor missing geographic coordinates (3 instances) were dropped.

Business rules were applied to ensure market representativeness: transactions below £50,000 and properties exceeding 10 habitable rooms or with areas under 15 m² were excluded. Outlier detection was performed using the Interquartile Range (IQR) method on the upper bounds of Price and *Total_Floor_Area* ($Q3 + 1.5 \times IQR$), removing extreme luxury anomalies. This yielded an initial pre-processed dataset of 547,948 transactions. A final quality filter was subsequently applied to remove records with anomalous or unclassifiable *CONSTRUCTION_Age_Band* values ('INVALID!', 'Unknown', '2042'), which are not meaningful predictive categories. This reduced the final analytical dataset to 514,345 market-representative transactions.

For Feature Engineering, spatial and temporal features were explicitly extracted: *Postcode_Outcode* was derived for spatial fixed effects, and *Year_of_Transfer* and *Month_of_Transfer* were engineered to capture seasonality and macroeconomic shifts. Prior to model training, standard categorical variables were encoded using One-Hot Encoding. Crucially, the target environmental feature, *Current_Energy_Rating*, was transformed via Ordinal Encoding (from A=7 to G=1) to preserve the inherent hierarchical structure of energy efficiency. For the OLS baseline model, continuous variables were standardised to ensure coefficient comparability. Tree-based ML models (Random Forest, LightGBM, and XGBoost) are scale-invariant by construction and were therefore trained on the unscaled feature matrix.

3.2.6 CANONICAL FEATURE SELECTION AND CROSS-MODEL PARITY ENFORCEMENT

Prior to hyperparameter optimisation, a unified and mathematically consistent feature space was established through a structured feature screening procedure. A baseline LightGBM model (500 estimators, GPU-accelerated, *random_state*=42) was trained on the full post-encoding feature matrix as a diagnostic instrument. LightGBM was selected for this screening role on account of its computational efficiency under GPU acceleration, which rendered a full-dataset baseline pass feasible within seconds, a critical advantage given the 514,345-transaction scale of the dataset.

Following training, the native split-frequency importance was extracted for every feature. Variables that registered exactly zero splits across all 500 trees were identified as having no predictive utility under the gradient boosting regime. In practice, these consisted exclusively of hyper-rare one-hot encoded dummy variables, predominantly construction age band combinations with near-zero frequency in the training partition, that the algorithm never found sufficiently discriminative to include in any decision node. These zero-contribution features were permanently removed from the feature matrix.

This procedure reduced the initial high-dimensional representation to a canonical 53-feature space, which was serialised and persisted to disk as a JSON artefact (*approved_features_53.json*). This file subsequently served as the single source of



truth for all competing model architectures: the Random Forest and XGBoost pipelines each loaded this canonical list at initialisation and filtered their feature matrices accordingly before any hyperparameter search was conducted. This "JSON lock" mechanism guarantees strict methodological parity across models, ensuring that all observed differences in predictive performance are attributable exclusively to algorithmic architecture rather than to variation in the input feature space.

It is acknowledged that frequency-based importance, as opposed to permutation or SHAP-based importance, carries a theoretical risk of discarding infrequently used but informative features. This risk is mitigated by the conservative threshold adopted: only features registering zero splits—features the algorithm declined to use under any configuration across 500 trees—were removed. Features with any non-zero contribution, regardless of their relative ranking, were retained.

With the canonical feature space thus established, the dataset was partitioned for model training and evaluation as described in the following section.

3.2.7 DATASET SPLITTING

To evaluate the generalisability of the predictive models and prevent overfitting, the refined dataset of 514,345 transactions was partitioned using a standard 80/20 train-test split strategy. Specifically, 80% of the data was allocated to model training and hyperparameter tuning, while the remaining 20% was strictly held out as an unseen test set for final model evaluation and XAI interpretation.

3.3 BASELINE MODEL: THE HEDONIC PRICE APPROACH (OLS)

To establish a robust methodological benchmark, this study employs a traditional HPM estimated via OLS regression. Rooted in Lancaster's consumer theory and Rosen's framework of implicit markets, the hedonic approach assumes that a property is a composite good. Its total price is determined by the implicit valuation of its individual utility-bearing characteristics (structural, spatial, and temporal). Estimating this baseline is crucial to quantify the added predictive value of transitioning to complex, non-linear ML algorithms.

Prior to model estimation, specific feature transformations were executed to comply with the strict mathematical assumptions of multiple linear regression:

- **Categorical Encoding & Multicollinearity Prevention:** Categorical features (such as *Property_Type*, *Construction_Age_Band*, and temporal variables) were transformed using One-Hot Encoding. To avoid perfect multicollinearity, commonly referred to as the "dummy variable trap", the `drop_first=True` parameter was strictly enforced, creating linearly independent reference categories. As noted in Section 3.2.3, the *Construction_Age_Band* variable was encoded in its raw, un-harmonised form, resulting in 27 binary dummy features; the rationale for this decision is detailed therein.



Ordinal Treatment of Energy Efficiency: Unlike standard categorical features, the target variable for the "green premium", *Current_Energy_Rating*, was processed using Ordinal Encoding. By assigning hierarchical numerical weights (where A = 7 through G = 1), the model preserves the inherent ordered structure of energy performance, allowing the regression coefficient to capture the marginal monetary effect of climbing the efficiency ladder.

A critical methodological distinction in this baseline model involves the treatment of spatial data. Standard OLS regression assumes a linear relationship between the independent variables and the target variable. Consequently, utilising continuous exact coordinates (Latitude and Longitude) would force the model to incorrectly assume that property prices infinitely increase or decrease in a straight line across the geographic plane.

To circumvent this limitation and capture spatial heterogeneity within a linear framework, geographic coordinates were excluded from the OLS model. Instead, spatial fixed effects were introduced via the *Postcode_Outcode* categorical variable. By converting these outcodes into dummy variables, the model successfully compartmentalises the study area's discrete sub-markets, absorbing hyper-local price variations without violating the linearity assumption.

The OLS model was trained on the 80% training partition, and its performance on the unseen 20% test set serves as the absolute baseline regarding predictive accuracy metrics (R^2 , MAE, RMSE, MAPE). Its inherent limitations in capturing complex, non-linear interactions, particularly regarding high-value luxury properties, provide the fundamental justification for the subsequent implementation of advanced ensemble learning methods.

3.4 ADVANCED PREDICTIVE MODELLING

To overcome the rigid linear constraints and spatial limitations of the baseline HPM, the analytical pipeline transitioned to advanced, non-linear ML architectures. This phase leverages the capacity of tree-based ensemble algorithms to automatically map complex, multi-dimensional interactions between structural features, environmental ratings, and property values without requiring pre-specified functional forms.

A fundamental methodological shift in the ML phase was the handling of geographic data. While the OLS model relied on categorical *Postcode_Outcode* dummy variables to prevent the violation of linearity assumptions, the tree-based ML models utilised continuous exact coordinates (Latitude and Longitude). Because decision trees operate through binary recursive partitioning, they can effectively isolate non-linear, hyper-local spatial boundaries, rendering the high-cardinality postcode dummies obsolete.



As established in Section 3.2.6, a preliminary LightGBM-based screening procedure reduced the initial feature matrix to a canonical 53-feature space, shared identically across all three ML pipelines to guarantee methodological parity.

Algorithm Selection and Hyperparameter Optimisation

The predictive modelling phase evaluated three leading tree-based ensemble architectures:

- **LightGBM (Light Gradient Boosting Machine):** An advanced *Boosting* framework utilising leaf-wise tree growth, renowned for its supreme computational speed and efficiency in handling large-scale tabular data (Equation 1).

$$Gain = \frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right] - \gamma \quad (1)$$

- **Random Forest:** A robust *Bagging* (Bootstrap Aggregating) algorithm designed to combat overfitting by training hundreds of deep, independent decision trees on random subsets of data and features, subsequently averaging their predictions to minimise variance (Equation 2).

$$\hat{y}_i = \frac{1}{B} \sum_{b=1}^B f_b(x_i) \quad (2)$$

- **XGBoost (eXtreme Gradient Boosting):** A highly sophisticated boosting algorithm that incorporates strict mathematical regularization (L1 and L2 penalties) to prevent overfitting, acting as the primary engine to capture the deepest non-linear market dynamics (Equation 3).

$$L(\phi) = \sum_i l(\hat{y}_i, y_i) + \sum_k \Omega(f_k) \quad (3)$$

where, $\Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2 + \alpha \|w\|_1$

To guarantee a scientifically fair and rigorous comparison between these algorithms, hyperparameter tuning was conducted using Bayesian Optimisation via the **Optuna** framework. Unlike rudimentary grid searches, Bayesian Optimisation utilises probabilistic models to intelligently navigate the hyperparameter space, converging on the optimal configuration with significantly less computational cost.

For each of the three ML algorithms, exactly 30 optimisation trials were executed to ensure parity in computational effort. Crucially, to satisfy the CRISP-ML mandate for total reproducibility, dual random seeds were strictly enforced. First, a Tree-structured Parzen Estimator (TPE) sampler with a fixed mathematical seed (seed=42) was implemented across all Optuna studies to control the hyperparameter exploration paths. Second, the internal random number generator for every ML algorithm was permanently locked (random_state=42) during both the trial phase and the final model training. These deterministic constraints ensure that the optimisation space remains identical upon replication, guaranteeing that the final predictive performance of the



models is solely a reflection of algorithmic superiority rather than stochastic chance. The best-performing model from this phase will subsequently serve as the foundation for the SHAP explainability framework.

3.5 EXPLAINABILITY AND EVALUATION METRICS

The final phase of the CRISP-ML framework involves rigorous quantitative evaluation and the extraction of interpretable economic insights. To establish the predictive superiority of the non-linear ML algorithms over the linear baseline, the out-of-sample performance on the 20% test set will be systematically quantified using four standard regression metrics:

- **Mean Absolute Error (MAE):** This metric provides a highly intuitive measure of predictive accuracy by calculating the average absolute difference between the predicted property prices and the actual transaction prices, expressed in British Pounds (£). It is robust to extreme outliers and reflects the average expected error per transaction (Equation 4).

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (4)$$

- **Root Mean Squared Error (RMSE):** By squaring the prediction errors before averaging them, RMSE heavily penalises large predictive discrepancies. In the context of the highly skewed UK housing market, a reduction in RMSE indicates a model's superior capability to accurately value high-end luxury properties without being distorted by their variance (Equation 5).

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (5)$$

- **Mean Absolute Percentage Error (MAPE):** This metric expresses the prediction error as a percentage of the actual property price, offering a relative measure of accuracy that allows for standardised comparison across different regional markets with varying base price levels (Equation 6).

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100 \quad (6)$$

- **Coefficient of Determination (R²):** The R² score quantifies the proportion of the variance in the dependent variable (property price) that is predictable from the independent variables, serving as the primary indicator of the model's explanatory power (Equation 7).

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (7)$$

3.5.1 MODEL INTERPRETABILITY VIA SHAP

Following the identification of the best-performing algorithm (the "Champion Model"), the research transitions from predictive modelling to explanatory analysis. Recognising the "black-box" nature of advanced tree ensembles, this study implements SHAP, a post-hoc XAI framework grounded in cooperative game theory. SHAP calculates the



exact marginal contribution of every feature to the final predicted price of each individual property (Equation 8).

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F|-|S|-1)!}{|F|!} [f_x(S \cup \{i\}) - f_x(S)] \quad (8)$$

It is important to emphasise that SHAP values represent model-based marginal contributions rather than causal effects. While they provide a consistent decomposition of the model's predictions, they do not imply that changes in a given feature would lead to equivalent changes in property prices in a real-world setting. As such, the estimated contributions of energy efficiency should be interpreted as conditional on the model and the observed data structure, rather than as strictly causal estimates.

Due to the extreme computational complexity required to compute exact Shapley values across deep tree structures for over 100,000 test set observations, a robust random sampling strategy was employed. A representative sub-sample of 50,000 properties was strategically extracted from the test data. This sample size is sufficiently large to guarantee statistical significance and accurately reflect macro-economic market dynamics while remaining computationally viable.

By isolating the SHAP values specifically assigned to the *Current_Energy_Rating* feature, this methodology enables the explicit extraction and visualisation of the exact financial premium (or discount) the market attributes to each energy efficiency band, completely decoupled from confounding variables such as spatial location or property size.

3.6 ETHICAL CONSIDERATIONS AND DATA PRIVACY

In compliance with academic and ethical standards for data science, this research relies exclusively on openly published Government Data (OGD). The Land Registry and EPC datasets are strictly anonymised at the source, containing no Personal Identifiable Information (PII) regarding the buyers or sellers. Furthermore, to mitigate algorithmic bias, a common ethical concern in real estate AVMs where black-box algorithms can inadvertently discriminate against certain demographic areas, this study explicitly integrates the SHAP framework. By enforcing model transparency, the research ensures that valuations are driven purely by structural, environmental, and objective spatial data, thereby upholding fairness and algorithmic accountability.



4. EMPIRICAL STUDY

This chapter details the empirical execution of the predictive modelling pipeline, transitioning from the foundational linear baseline to the advanced ML architectures. The primary objective of this phase is to evaluate and identify the optimal algorithm for property valuation in the UK market, which will subsequently serve as the analytical engine for the SHAP explainability framework in Chapter 5.

4.1 BASELINE ESTABLISHMENT (THE OLS PERFORMANCE)

To empirically quantify the added value of complex algorithmic approaches, a traditional OLS regression was first executed on the training partition. Acting as the study's benchmark, the OLS model utilised One-Hot Encoded categorical variables and relied on postcode Outcodes as spatial fixed effects to circumvent the linearity violations associated with continuous geographic coordinates.

The out-of-sample evaluation on the 20% test set revealed the inherent limitations of traditional econometrics when applied to dynamic, multi-regional housing markets. The OLS model achieved a baseline Coefficient of Determination (R^2) of 0.7458, indicating a reasonable, yet incomplete, capture of market variance. However, the magnitude of the errors exposed significant predictive limitations. The model registered an MAE of £79,449 and RMSE of £114,401.

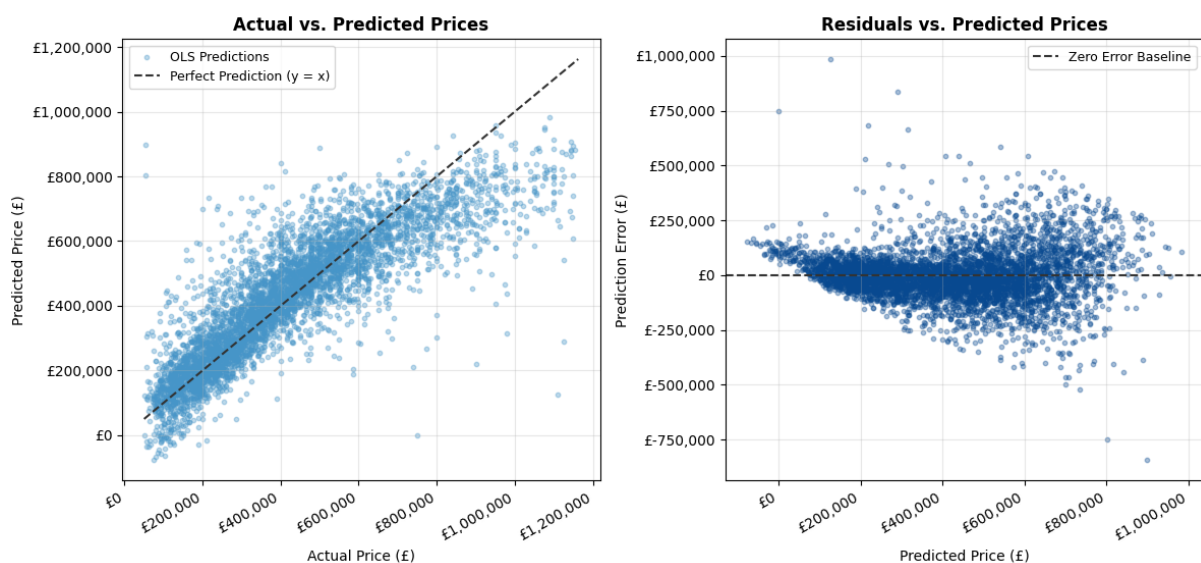


Figure 4.1 - Baseline Model Performance and Error Distribution

The substantial gap between the MAE and RMSE suggests that the linear model struggled profoundly with high-value property outliers, failing to capture the non-linear price appreciation typical of the luxury segment. This predictive limitation is visually corroborated in Figure 4.1, which illustrates the actual versus predicted prices alongside the residual distribution. The scatter plots reveal a distinct heteroscedastic pattern, where the prediction error widens significantly as property values increase.



Furthermore, the Mean Absolute Percentage Error (MAPE) settled at 24.82%, a high margin of relative error that underscores the inadequacy of relying solely on categorical postcode boundaries to capture continuous, hyper-local spatial dynamics.

4.2 MACHINE LEARNING EXECUTION AND HYPERPARAMETER TUNING

Following the baseline establishment, the analytical focus shifted to tree-based ensemble methods: Random Forest, LightGBM, and XGBoost. Unlike the OLS model, these algorithms were trained using continuous exact coordinates (Latitude and Longitude), allowing the recursive partitioning mechanisms to automatically delineate hyper-local market boundaries.

During the hyperparameter optimisation phase, the two Boosting architectures (LightGBM and XGBoost) demonstrated rapid and highly efficient convergence. Leveraging GPU acceleration, the optimisation framework successfully mapped their optimal configurations, fine-tuning critical parameters such as learning rates, maximum tree depth, and subsampling fractions. Both boosting models inherently capitalised on the canonical 53-feature space during their sequential tree generation, allowing them to aggressively capture non-linear spatial and structural interactions right from the initial trials.

Conversely, a critical algorithmic behaviour regarding spatial data was observed and corrected in the Random Forest architecture. In its default configuration, the Random Forest relies on a square root heuristic (`max_features='sqrt'`) for feature selection at each node split while operating within the canonical 53-feature space (established in Section 3.2.6), which restricted the algorithm to evaluating only approximately 7 variables per split, frequently blinding the decision trees to crucial spatial coordinates. To rectify this, the hyperparameter search space was expanded to allow the algorithm to evaluate higher fractions of features per split (0.6, 0.8, and 1.0), alongside standard parameters such as tree depth and minimum leaf samples. The Bayesian optimisation ultimately converged on an optimal `max_features` fraction of 0.8. This strategic intervention restored the algorithm's spatial awareness, drastically reducing its RMSE from an initial £115,503 to £96,610, empirically demonstrating the importance of custom hyperparameter definition when modelling continuous spatial data.

4.3 COMPARATIVE MODEL EVALUATION

With all models optimised and trained, a definitive comparative analysis was conducted on the unseen test set. The comprehensive predictive performance across the four evaluated metrics is detailed in Table 4.1. This comparative data is visualised in Figure 4.2, which provides a definitive overview of the performance of all tested algorithms across the metrics of Explanatory Power (R^2), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE).



Table 4.1 - Predictive Performance of the Baseline and Machine Learning Models on the out-of-sample test set

Model / Metric	R ²	MAE	RMSE	MAPE
OLS (Baseline)	0.7458	£ 79,449.17	£ 114,401.78	24.82%
Random Forest	0.8187	£ 62,201.62	£ 96,610.38	18.16%
LightGBM	0.8233	£ 61,545.29	£ 95,386.78	17.96%
XGBoost (Champion)	0.8249	£ 60,868.46	£ 94,954.19	17.67%

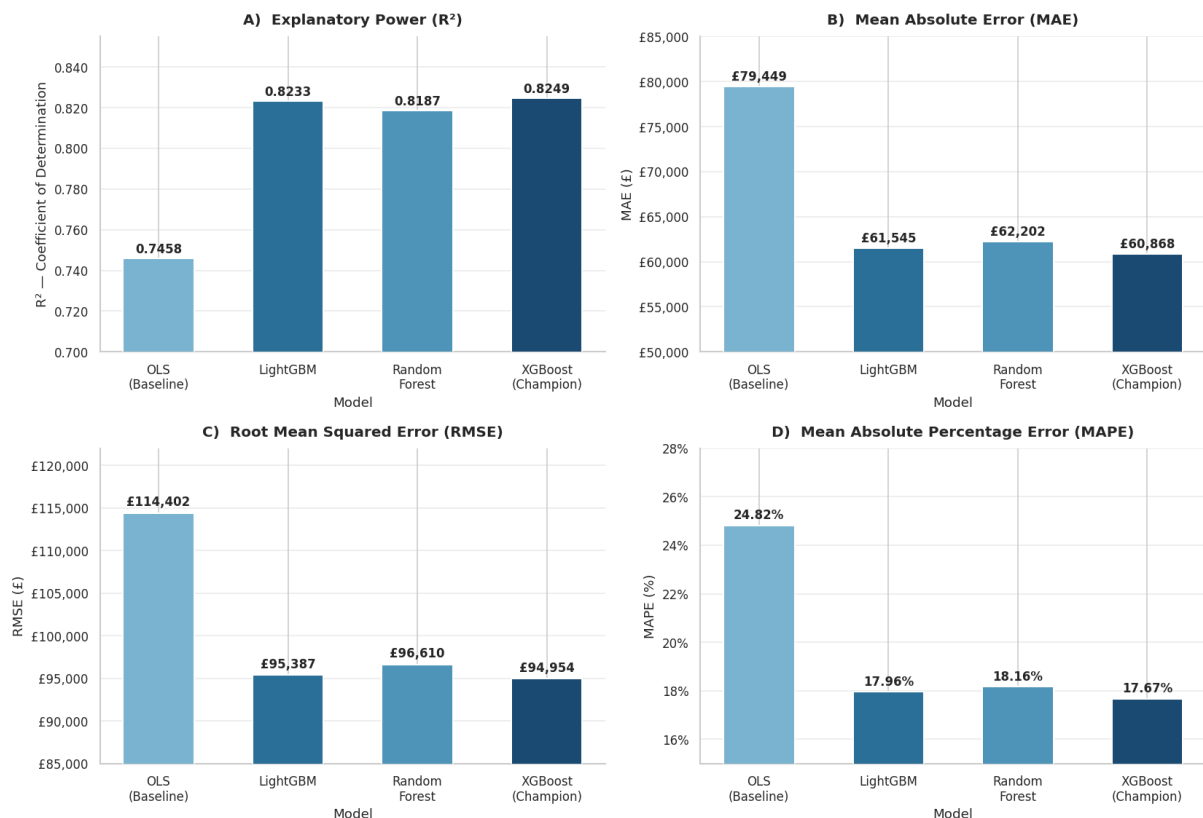


Figure 4.2 - Global Performance Metrics Across Models

The transition from the linear baseline to non-linear tree ensembles generated immediate predictive disruption. The Bagging architecture, represented by the optimised Random Forest, successfully bypassed the £100k error threshold, achieving an R² of 0.8187 and reducing the MAE to £62,201. Furthermore, the LightGBM demonstrated exceptional computational efficiency and formidable precision, reaching an R² of 0.8233.

Ultimately, XGBoost emerged as the absolute champion of the empirical study. By achieving the highest R² (0.8249) and minimising the average absolute error to £60,868, the XGBoost model successfully reduced the average prediction error by £18,581 per transaction when compared to the OLS baseline. Its MAPE of 17.67% confirms its robustness across the entire price distribution.



The scatter plot visualisation of the XGBoost predictions in Figure 4.3 further corroborates its superiority.

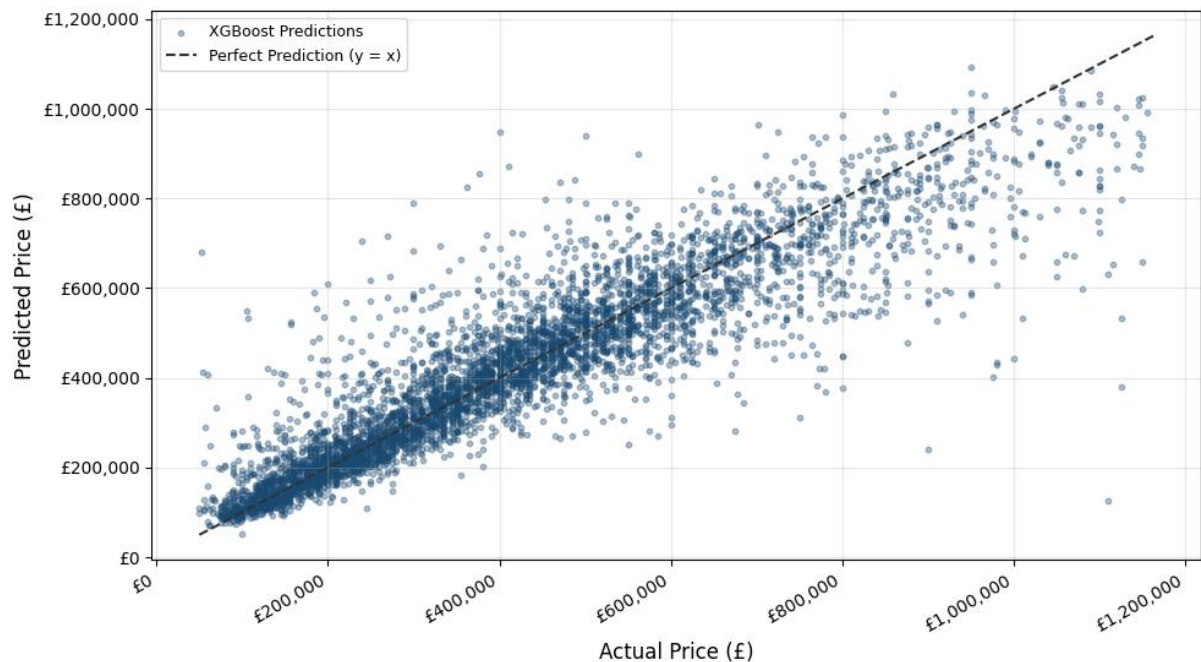


Figure 4.3 - Actual vs. Predicted Property Prices (XGBoost)

The dense clustering of predictions along the dashed line of perfect fit, particularly within the £100,000 to £600,000 range, visually validates the algorithm's capacity to internalise complex non-linear interactions and overcome the heteroscedasticity that plagued the linear baseline. (Corresponding visualisations for the Random Forest and LightGBM models are also available in Appendix A for comprehensive reference). With XGBoost confirmed as the optimal predictive architecture, the analysis now turns to its central scientific objective: decomposing that predictive signal into exact monetary attributions for every feature, at the level of the individual property.



5. RESULTS AND DISCUSSION

The empirical study conducted in Chapter 4 established XGBoost as the most robust and accurate Automated Valuation Model (AVM) for the UK housing market. To overcome the inherent opacity of these advanced algorithms and answer the core research question of this thesis, it is imperative to transition from pure prediction to model explainability.

Operationalising the TreeSHAP methodology described in Section 3.5.1, this chapter deconstructs the XGBoost predictions at the individual property level. By doing so, this section isolates the spatial, structural, and environmental dynamics of the market, with a primary focus on quantifying the "Green Premium" and "Brown Discount" associated with Energy Performance Certificates (EPC).

5.1 OPENING THE BLACK-BOX: SHAP GLOBAL INTERPRETABILITY

Before isolating the specific impact of energy efficiency, it is crucial to validate the overall economic logic internalised by the algorithm. A model cannot accurately price a "Green Premium" if it fails to comprehend the foundational rules of real estate valuation, namely the paramount importance of location and property size.

To render the model's internal reasoning interpretable and monetarily quantifiable, this chapter employs SHAP, computed via the TreeSHAP algorithm (Lundberg et al., 2020). Formally rooted in cooperative game theory, SHAP treats each feature as a player in a coalition and assigns each player a "Shapley value", representing its fair share of the total prediction, calculated by averaging its marginal contribution across all possible orderings of the feature set. In the present context, because the target variable is denominated in pounds sterling, every SHAP value is directly interpretable as a monetary quantity. For any given property, the sum of all feature SHAP values equals the difference between the model's individual prediction and the baseline value, which is defined as the global mean predicted price (£412,950.188, corresponding to the model's baseline intercept). A positive SHAP value of, for instance, +£13,506 for a Band A EPC rating signifies that the model attributes an increase of £13,506 in predicted price principally to that property's energy rating, holding all other covariates constant in the Shapley sense. A negative value denotes a corresponding price reduction. All monetary SHAP values reported throughout Chapter 5 derive from this definition and should be interpreted accordingly.

Figure 5.1 presents the SHAP Summary Plot (Beeswarm), which aggregates the Shapley values across the test set. This visualisation ranks variables by their global impact, where the horizontal displacement indicates the magnitude of a feature's contribution to the price, and the colour represents the feature value (red for high, blue for low).

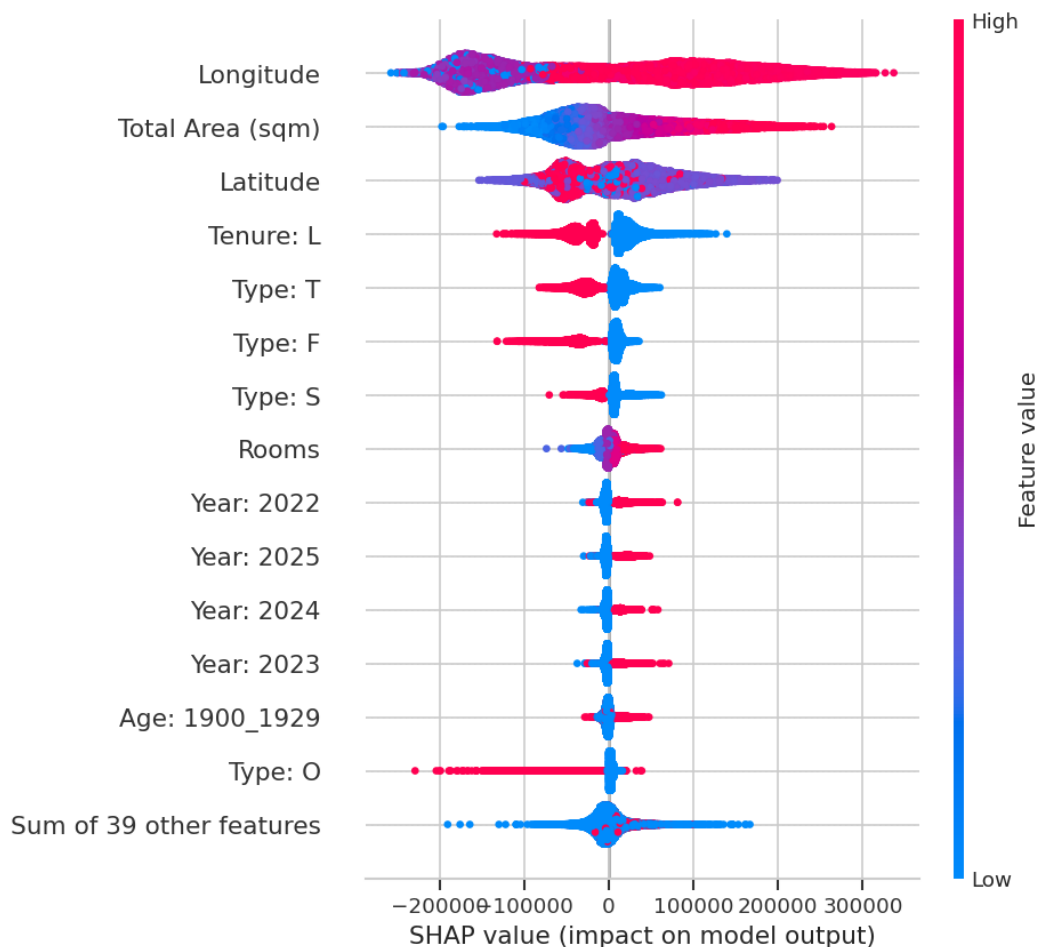


Figure 5.1 - Global Feature Importance (SHAP Summary Plot for the XGBoost Model)

As theoretically anticipated, the XGBoost model identified spatial coordinates and structural dimensions as the primary determinants of property value. **Longitude** and **Latitude** dominate the top of the hierarchy, reflecting the "Location, Location, Location" axiom. The plot reveals that geographic positioning can impose extreme penalties, down to -£200,000, or massive hyper-local premiums exceeding +£300,000.

The *Total Area (sqm)* emerged as the critical structural driver. The distribution shows a perfect correlation with market mechanics: high feature values (red dots) stretch drastically to the right, adding up to £250,000 to the baseline valuation, while smaller properties (blue dots) cluster on the left, consistently subtracting value. This confirms the algorithm's capacity to internalise the non-linear relationship between size and price.

Regarding **Property Types**, the SHAP summary plot reveals a consistent discount for high-density housing types relative to the model's baseline (the average predicted market price). For instance, properties identified as *Type_F* (Flats) or *Type_T* (Terraced houses), represented by red clusters, exert significant negative pressure on the valuation. Conversely, the absence of these characteristics (represented by blue



clusters) acts as a positive driver, as the model associates the exclusion of lower-priced typologies with a higher probability of the asset being a higher-value detached or semi-detached dwelling. The *Type_O* (Other) category displays a high variance, reflecting the heterogeneous nature of non-standard dwellings, which can either severely penalise or occasionally increase the valuation depending on their specific characteristics.

A similar logic applies to *Tenure_L*, where the model identifies a clear penalty for leasehold properties, aligning with the traditional market preference for freehold ownership. Conversely, the temporal variables demonstrate the effect of market appreciation over time. For instance, the red clusters for *Year_2025* are positioned to the right of the neutral axis, indicating that transactions occurring in later years receive a positive price adjustment, effectively capturing the inflationary trends of the 2021-2025 period.

Regarding the primary focus of this research, the energy efficiency indicators (such as *Current_Energy_Rating*) do not appear in the top 14 macro-drivers, being aggregated within the **"Sum of 39 other features"**. From an economic standpoint, this behaviour is expected: sustainability ratings do not dictate the baseline value of an asset in the same magnitude as location or square footage. Instead, the "Green Premium" acts as a marginal gain, a secondary layer of value. Consequently, to accurately quantify this environmental margin and isolate it from structural noise, a granular microeconomic analysis via SHAP dependence plots is required in the following section.

5.2 THE "GREEN PREMIUM": ISOLATING THE IMPACT OF ENERGY EFFICIENCY

Having established the macroeconomic drivers of the housing market and defined the SHAP monetary attribution framework in Section 5.1, the analysis shifts to the primary objective of this research: extracting and quantifying the non-linear financial impact of energy efficiency because the *Current_Energy_Rating* acts as a marginal gain rather than a baseline price determinant, its true economic value cannot be accurately assessed in a global summary plot.

By exclusively isolating the *EPC_Rating* variable through a SHAP Dependence Plot, macroeconomic "noise" (such as spatial location and total floor area) is removed. This methodology allows for the precise quantification of the financial premium or penalty the market attributes to each energy efficiency tier. The extracted mean SHAP values reveal the undeniable existence of both a substantial "Green Premium" and a severe "Brown Discount" within the UK market, as shown in Table 5.1.

The "Green Premium" (The Peak of Efficiency): Reaching the highest tier of energy efficiency (Band A) translates into an overwhelming financial premium. On average, the XGBoost model attributes a direct increase of £ 13,506.04 to the transaction value of these highly efficient properties. Band B also presents a consolidated positive premium of +£3,366.37, validating the market's willingness to pay for modern, sustainable infrastructure.



The Market's "Absolute Zero" (Level D): The intermediate levels hover near the axis of neutrality. Notably, Band D, which represents the most prevalent efficiency tier in the analytical dataset (as evidenced in Table 3.3) and the historical national average, presents a marginal impact of exactly +£54.96. This near-zero value perfectly corroborates economic theory: the market has fully internalised the D rating as the absolute standard, offering neither a premium nor demanding a significant discount.

The "Brown Discount" (The Penalty for Extreme Inefficiency): On the opposite end of the spectrum, severely energy-obsolete properties suffer profound depreciation. A Band G certification (the lowest legal level) imposes an average penalty of -£12,017.23 on the final property price. This demonstrates that the market punishes extreme inefficiency almost symmetrically to how it rewards structural excellence.

Table 5.1 - Mean Impact per EPC Band in £

EPC Band	Mean Impact (£)
G	-12,017.23
F	-428.12
E	933.07
D	54.96
C	-1,669.16
B	3,366.37
A	13,506.04

A notable anomaly merits explicit discussion. Band C exhibits an apparently paradoxical negative average SHAP contribution of -£1,669.16, despite representing above-average energy efficiency relative to the market's neutral baseline of Band D (+£54.96). This non-monotonic outcome reflects a structural "no man's land" in the residential valuation hierarchy. Properties that attain Band C frequently belong to the mid-century post-war architectural stock (circa 1950–1980) that has undergone partial, incremental retrofitting. Crucially, these assets possess neither the high-efficiency infrastructure of modern Band A/B new-build properties that commands the pure Green Premium, nor the heritage authenticity of pre-1900 architecture that triggers the Historical Forgiveness effect documented in Section 5.2.1. Consequently, relative to the global mean, they attract a modest depreciation.

Importantly, this pattern is not a sampling artefact: the Band C penalty is consistent across all three regional markets examined in Section 5.3, registering -£2,233.03 in Cornwall, -£1,598.11 in Greater London, and -£1,704.45 in Greater Manchester, confirming that the anomaly reflects a genuine and systematic market behavioural regularity. It should further be noted that Band E records a marginally positive contribution (+£933.07) despite representing below-average efficiency. Much like the extremes of the lower bands, this can be attributed to a partial heritage effect: many older properties with high architectural character fall into this tier, where historical



premiums begin to incrementally offset the expected energy-related discounts. This renders the full EPC valuation scale non-monotonic at both the C and E positions (Figure 5.2), an empirical complexity that a linear HPM is wholly unable to recover.

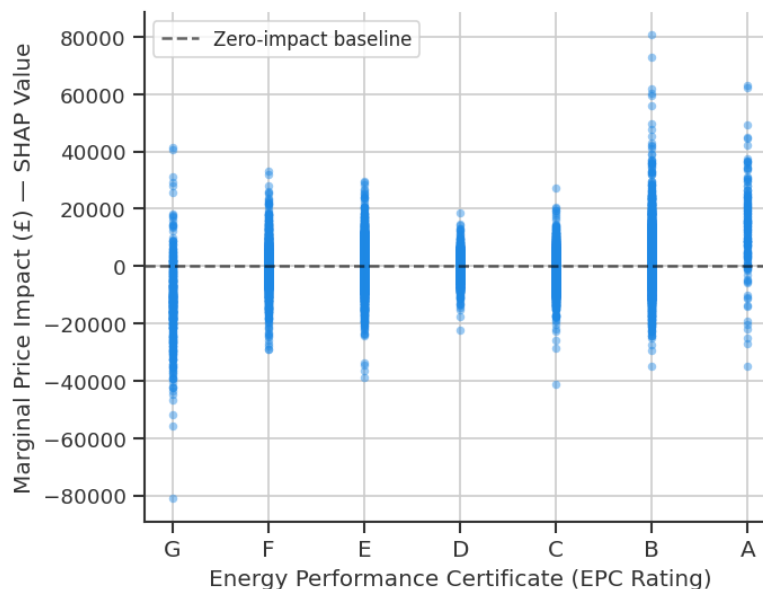


Figure 5.2 - SHAP Dependence Plot: Marginal Impact of the Energy Certificate (£)

5.2.1 INTERSECTIONAL DYNAMICS: THE HISTORICAL "FORGIVENESS" EFFECT

While the absolute extremes (Bands A and G) follow a logical economic trajectory, the mean SHAP values for the intermediate-low tiers (Bands E and F) exhibit significant variance and unexpected averages (e.g., Band E displaying a slight positive average). This conditional variance suggests the presence of strong interaction effects. As hypothesized during the Exploratory Data Analysis phase, a property's construction age acts as a critical confounding variable when evaluating energy obsolescence.

To deconstruct this dynamic, a SHAP Interaction Plot was generated, isolating the marginal impact of the energy rating while differentiating historical properties (built before 1900, representing Victorian and Edwardian architecture) from the rest of the housing stock.

The visualisation depicted in Figure 5.3 reveals a highly sophisticated market behaviour regarding the "Brown Discount". In the lowest efficiency tiers (Certificates G, F, and E), the price impact exhibits extreme vertical dispersion, ranging from severe penalties to positive marginal premiums. Crucially, a significant cluster of historical properties is positioned above the neutrality line. This demonstrates that the market actively "forgives" energy inefficiency if the asset possesses historical, architectural, or period-specific charm. For these properties, the inherent heritage value effectively neutralizes, and sometimes overrides, the anticipated "Brown Discount."

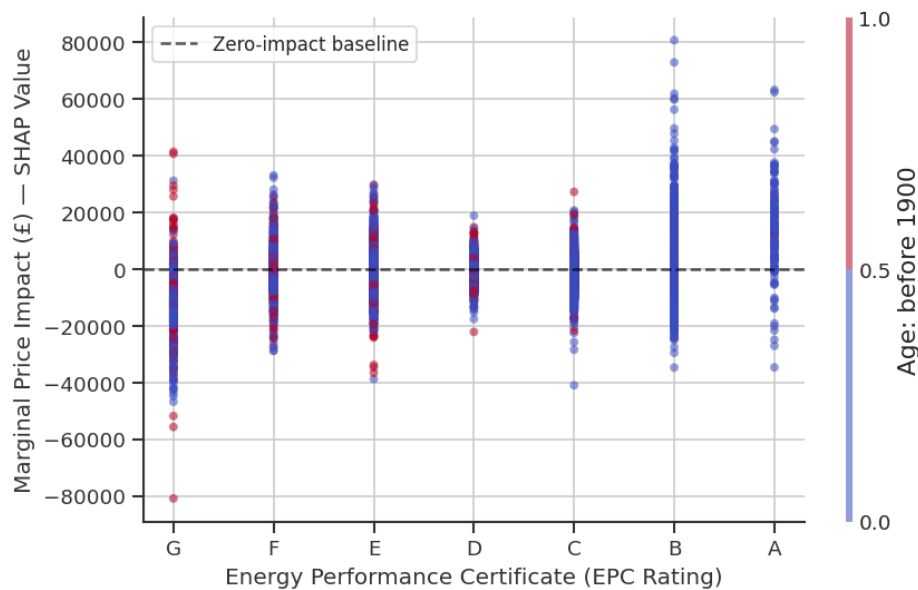


Figure 5.3 - SHAP Interaction Plot: Green Premium vs. Historical Age

Conversely, the deepest absolute financial penalty in the G band is also associated with a historical property (represented by the isolated red dot approaching -£80,000). The model dictates a bifurcated reality for historical assets: while some are 'forgiven' due to their cultural premium, others are ruthlessly penalised for their energy obsolescence. This severe discount perfectly captures the 'retrofitting debt', the massive anticipated capital expenditure and regulatory complexity required to bring 19th-century Victorian architecture up to modern efficiency standards

5.2.2 LOCAL INTERPRETABILITY: DECONSTRUCTING EXTREME OUTLIERS

While the SHAP Interaction Plot (Figure 5.3) provides a macro-level perspective on how architectural era mitigates or amplifies the Brown Discount across the full distribution, the TreeSHAP framework permits an equally rigorous diagnosis at the level of the individual transaction. Within the SHAP distribution for energy efficiency, the model identified five properties that occupy the absolute statistical extremes: one property suffered a severe penalty approaching -£81,000; one Band G property defied conventional expectation by commanding a positive energy efficiency contribution of +£41,317; and three distinct assets captured premiums of approximately +£81,000, +£73,000, and +£63,000 respectively. By isolating these extremes, it is possible to observe precisely how the XGBoost model evaluates the complex intersection of energy efficiency, physical scale, architectural era, and geography, interactions that a linear OLS model is constitutionally incapable of capturing.

1. The Extreme "Brown Discount" (-£80,899)

The waterfall plot (Figure 5.4) deconstructs the most severely penalised property in the test set. Departing from the global mean predicted price of approximately £412,950, the valuation is driven substantially upward by the property's central London location



(+£150,765), its substantial floor area of 132.0 sqm (+£88,731), the freehold ownership of the asset (+£26,347), and the heritage premium associated with its Victorian construction era (+£10,376). This confluence of positive attributes establishes an exceptionally high predicted baseline.

However, the Band G energy rating is associated with a single dominant deduction of –£80,899 that nearly eliminates the entire size premium in one stroke (Table 5.2). The model has effectively priced in the full magnitude of this property's "retrofitting debt": the immense capital expenditure, specialist conservation labour, complex planning permissions and potential listed building constraints, and the sustained exorbitant energy costs associated with bringing a large pre-1900 terraced house in central London to modern efficiency standards. Crucially, the property's Victorian construction era, which confers a positive heritage premium of +£10,376, simultaneously amplifies the Brown Discount by increasing the anticipated cost and regulatory complexity of retrofitting, a duality that the non-linear XGBoost model captures with precision and that a fixed-coefficient OLS estimator would collapse into a single averaged coefficient.

Table 5.2 - Profile of the Most Penalised Property (Index #296154)

Feature	Value
County	Greater London
EPC Rating	G
Total Floor Area	132.0 sqm
Habitable Rooms	5
Property Type	Terraced
Tenure	Freehold
Construction Era	Pre-1900
Date of Transfer	Apr-23
EPC SHAP Impact (£)	£-80,899.89

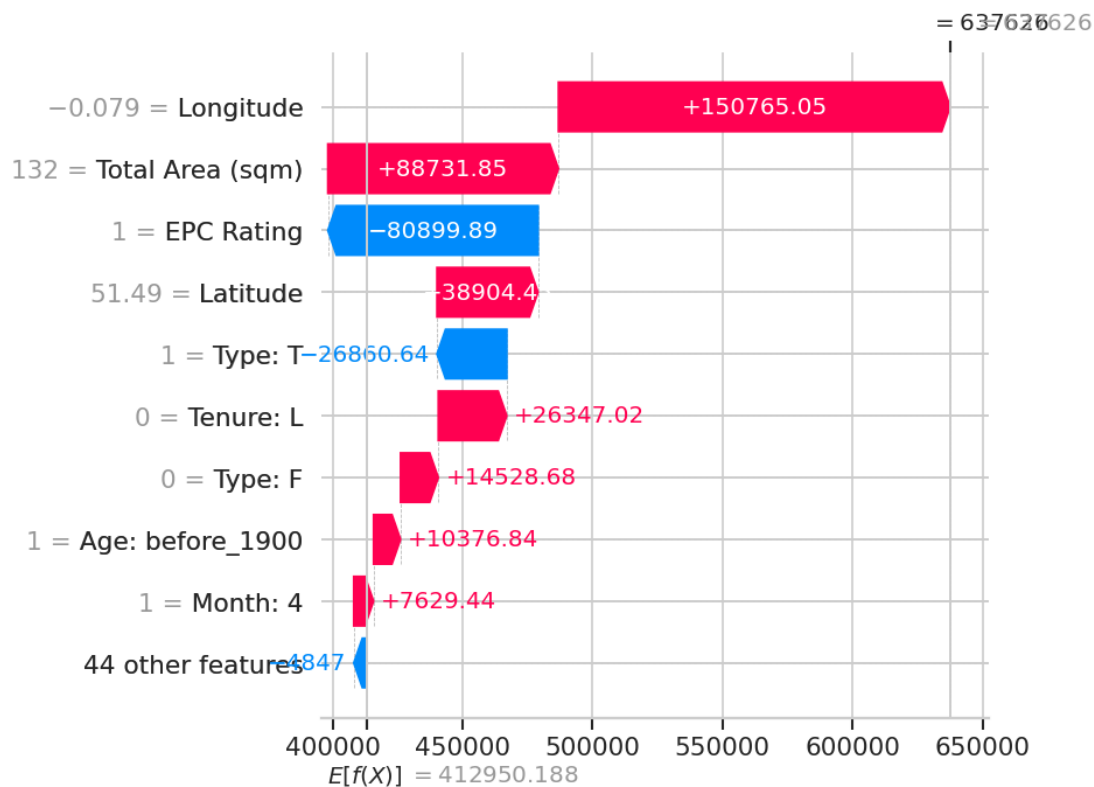


Figure 5.4 - Extreme 'Brown Discount' for Property #296154

2. The Historical Forgiveness Anomaly (+£41,317): When Band G Commands a Premium

The most theoretically significant finding in the local interpretability analysis is not the most severely penalised property, but the most anomalous one: a Band G property that commands a positive EPC SHAP contribution of +£41,316.78. This result constitutes the most direct empirical proof, at the level of the individual transaction, of the "Historical Forgiveness" effect identified at the aggregate level in Section 5.2.1.

The waterfall plot (Figure 5.5) reveals the structural logic of this extreme outcome. Property #31531 is a large, detached house of 138.6 sqm with five habitable rooms, constructed before 1900 and held under freehold tenure in Cornwall, transacted in March 2021. Its geographical coordinates place it within the rural Cornish interior, a market characterised by a pronounced premium for physical scale, privacy, and period architectural authenticity, as established in the regional analysis of Section 5.3. In this specific market context, the model has learned a relationship that no linear HPM could capture: for a property of this architectural and geographical profile, the absence of modern retrofitting is not merely forgiven, it is actively rewarded.

The EPC SHAP impact of +£41,316.78 represents a deviation of +£53,334 above the national average Band G penalty of -£12,017.23, reversing its sign entirely (Table 5.3). The algorithm's interpretation is economically coherent. Buyers in the market for a large pre-1900 Cornish detached property are self-selecting for architectural



authenticity and rural heritage. In this buyer segment, the original fabric of the building—the stone walls, solid floors, and unaltered structural character that are simultaneously responsible for its poor thermal performance—is not a liability to be remediated but an asset to be preserved. The market effectively prices the absence of modern energy intervention as evidence that the property's heritage integrity is intact, assigning a premium that completely overwhelms and reverses the standard energy efficiency penalty. This is the Historical Forgiveness effect at its most extreme: the model has successfully internalised that for a specific class of pre-industrial Cornish assets, a Band G certificate is not a deficiency but a mark of architectural purity.

Table 5.3 - Profile of the Historical Forgiveness Outlier (Index #31531)

Feature	Value
County	Cornwall
EPC Rating	G
Total Floor Area	138.6 sqm
Habitable Rooms	5
Property Type	Detached
Tenure	Freehold
Construction Era	Pre-1900
Date of Transfer	Mar-21
EPC SHAP Impact (£)	+£41,316.78

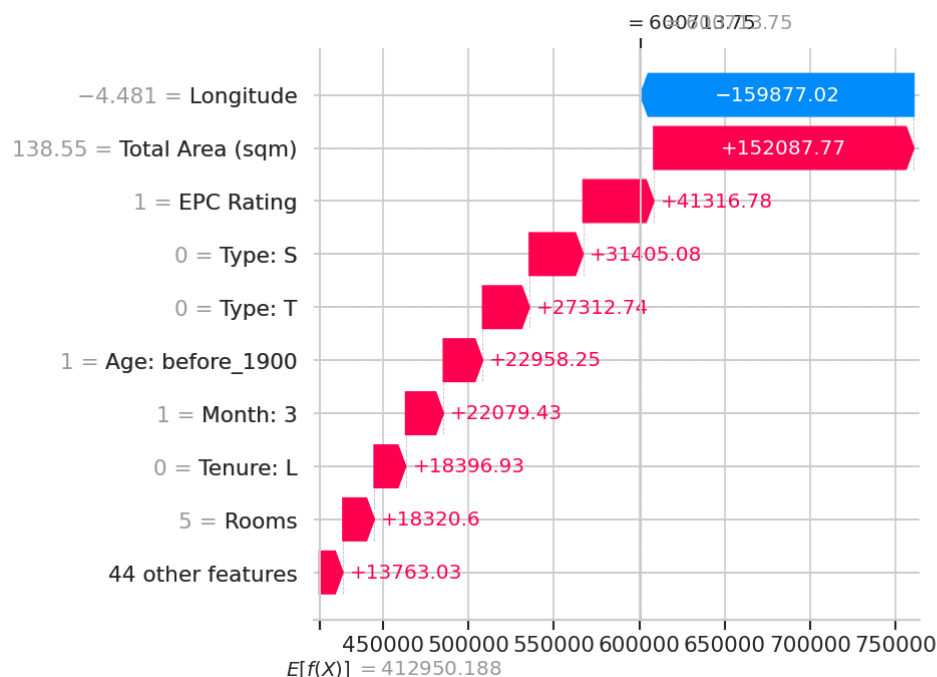


Figure 5.5 - Waterfall Plot: Historical Forgiveness Anomaly, Property #31531



3. A Cross-Regional Case Study: The Top 3 "Green Premiums"

Conversely, analysing the structural DNA of the three properties that received the highest energy efficiency premiums reveals a fascinating geographical and architectural diversity. This proves that the premium is highly elastic and not just a flat rate:

Rank 1: The Coastal Micro-Retreat (Cornwall | +£80,596)

The waterfall plot (Figure 5.6) for the highest absolute premium reveals a distinct valuation dynamic. Starting from the £412,950 average, this property's remote Cornwall location (Longitude) and tiny 36 sqm scale initially impose severe market penalties (-£72,589 and -£45,697, respectively). However, its Freehold status (+£49,543) and its **Band B** energy rating completely reverse this trajectory (Table 5.4). The EPC rating emerges as the single largest positive value driver, injecting a staggering +£80,596. In a coastal, leisure-driven market, highly efficient micro-properties (such as holiday lets) are exceptionally attractive. The model rewards this efficiency as it minimises operational costs, directly protecting rental yields and completely offsetting the traditional penalties for small size and remote location.

Table 5.4 - Profile of the Top 1 Premium Property (Index #402155)

Feature	Value
County	Cornwall
EPC Rating	B
Total Floor Area	36.0 sqm
Habitable Rooms	3
Property Type	Other
Tenure	Freehold
Construction Era	1983–1990
Date of Transfer	Jun-24
EPC SHAP Impact (£)	+£80,596.88

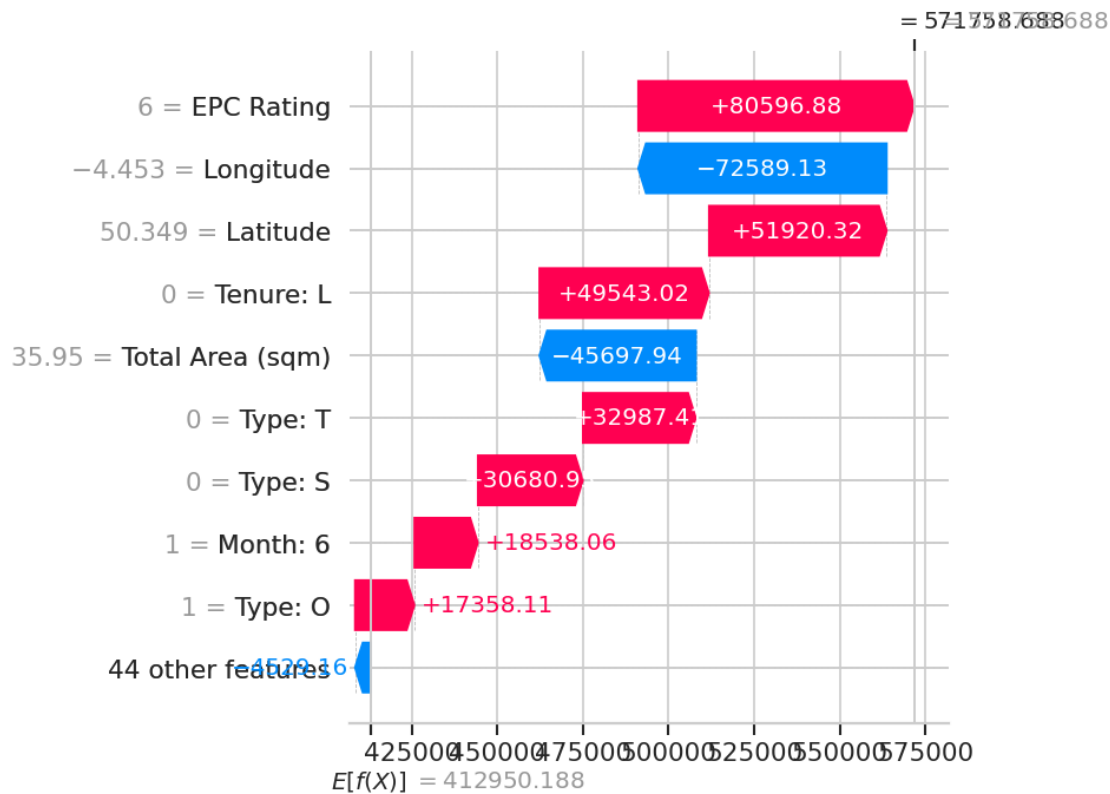


Figure 5.6 - Extreme “Green Premium” (Rank 1) for Property #402155

Rank 2: The Historical Conversion (Greater Manchester | +£72,786)

The waterfall plot (Figure 5.7) for the second-highest premium (+£72,786) illustrates a remarkable market dynamic. Starting from the £412,950 baseline, this property suffers a massive locational penalty (-£147,275 due to Longitude) and a scale penalty (-£39,442 for its small 47 sqm area). However, the model heavily rewards its Freehold status (+£40,359) and, crucially, its **Band B** rating, which acts as the primary value recovery mechanism (+£72,786) (Table 5.5). Achieving such high efficiency in an early 20th-century building is technically challenging. The algorithm recognises a "turnkey" historic asset, financially rewarding the fact that the complex, capital-intensive retrofitting required to merge period character with modern energy standards has already been completed.



Table 5.5 - Profile of the Top 2 Premium Property (Index #301818)

Feature	Value
County	Greater Manchester
EPC Rating	B
Total Floor Area	47.0 sqm
Habitable Rooms	2
Property Type	Other
Tenure	Freehold
Construction Era	1900–1929
Date of Transfer	May-23
EPC SHAP Impact (£)	+£72,786.91

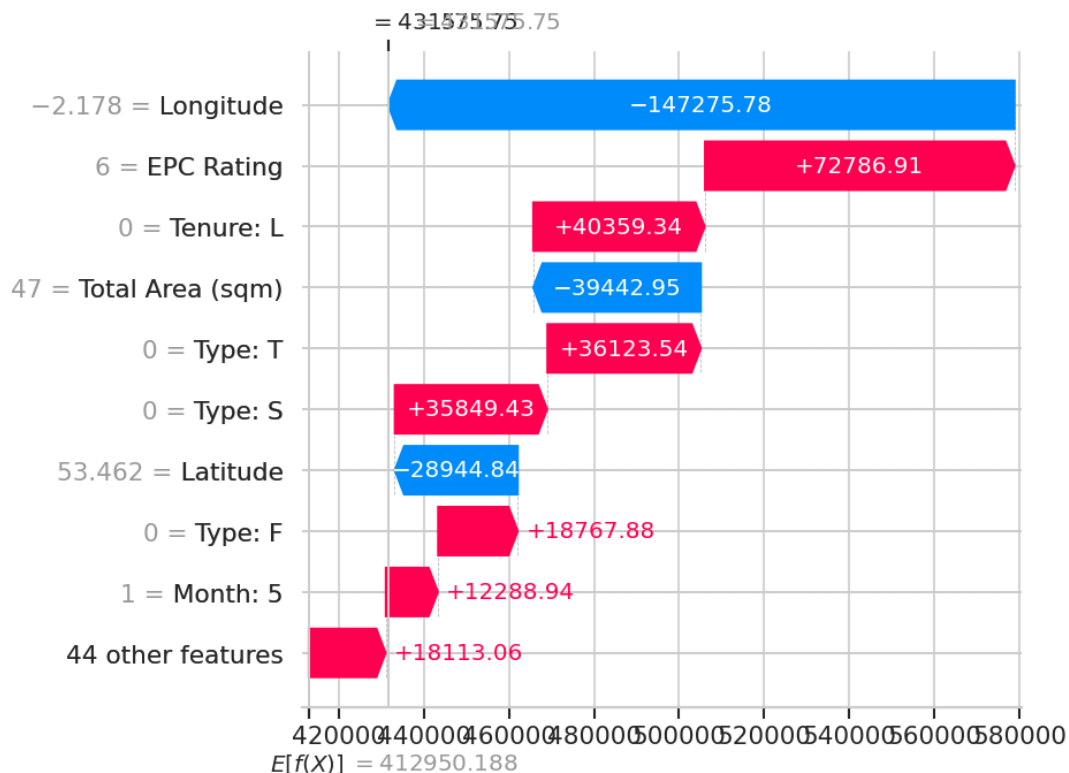


Figure 5.7 - Extreme “Green Premium” (Rank 2) for Property #301818

Rank 3: The New-Build Coastal Family Home (Cornwall | +£63,111)

The waterfall plot (Figure 5.8) for the third-ranking premium presents the story of a property rewarded by the model across virtually every dimension simultaneously, contending with a single dominant geographical penalty. Departing from the global mean of approximately £412,950, the substantial floor area of 118.0 sqm generates a positive contribution of +£123,336, the freehold ownership of the asset contributes +£21,309, and the detached typology injects combined architectural premiums exceeding +£65,000, reflecting the considerable market premium that a self-contained



dwelling commands over terraced and semi-detached typologies. Against this portfolio of positive contributors, the property's remote Cornish location imposes a penalty of -£90,569, reflecting the market discount applied to the far south-west peninsula relative to higher-density urban and suburban markets. It is the Band A energy certificate that provides the critical final contribution of +£63,111, functioning as the value driver that substantially offsets this locational discount. For a large detached family home where baseline space-heating demands are intrinsically high, achieving the highest tier of energy efficiency translates directly into material long-term reductions in operating costs, which the market prices at over £63,000, demonstrating that at scale and in conjunction with a detached typology (Table 5.6), the Green Premium amplifies substantially beyond the national average of +£13,506.

Table 5.6 - Profile of the Top 3 Premium Property (Index #202067)

Feature	Value
County	Cornwall
EPC Rating	A
Total Floor Area	118.0 sqm
Habitable Rooms	4
Property Type	Detached
Tenure	Freehold
Construction Era	2017
Date of Transfer	Jun-22
EPC SHAP Impact (£)	+£63,111.25

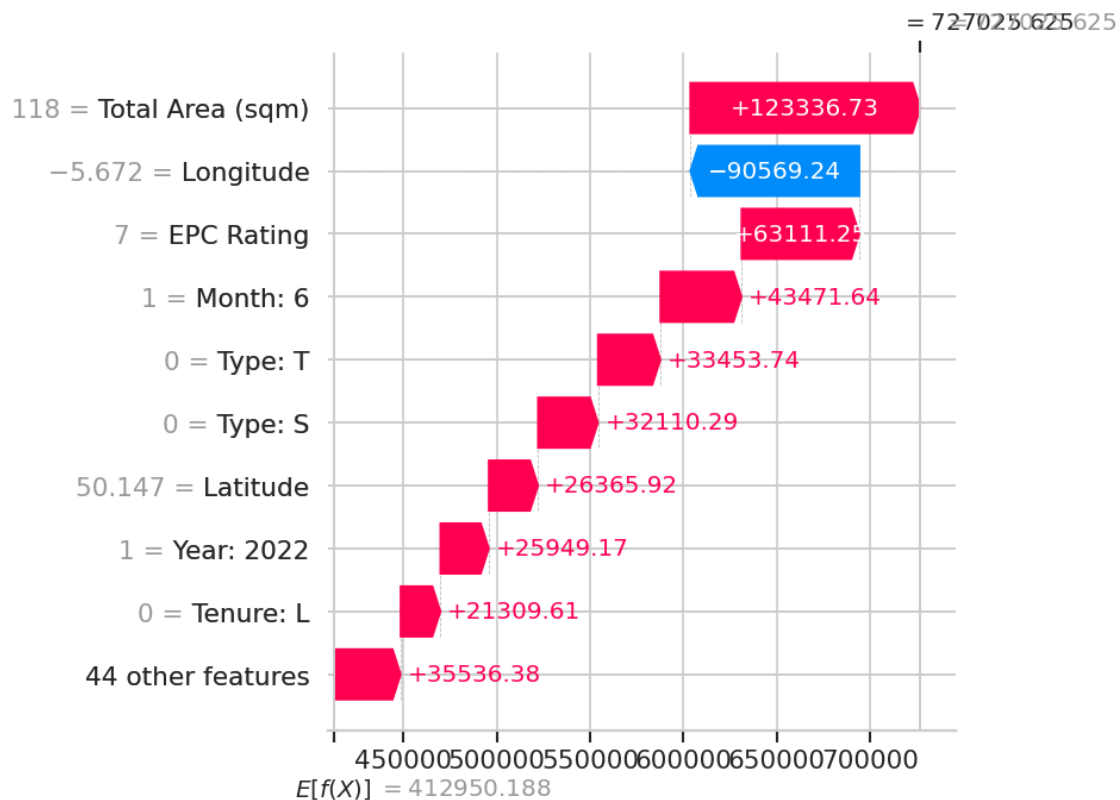


Figure 5.8 - Extreme “Green Premium” (Rank 3) for Property #202067

These case studies mathematically demonstrate that the market valuation of energy efficiency is profoundly contextual: the premium scales non-linearly with physical size, architectural typology, and geography. As these extreme examples clearly show, regional context plays a pivotal role in pricing, seamlessly leading to a broader analysis of spatial dynamics.

5.3 SPATIAL AND STRUCTURAL DYNAMICS

To deepen the economic validity of the model, it is necessary to test the hypothesis of spatial heterogeneity. The theoretical premise suggests that the valuation of energy efficiency and the fundamental structural drivers of price are not uniform across the British territory. To validate this, the exact SHAP values were cross-referenced with their respective regional markets: Greater London, Greater Manchester, and Cornwall.

5.3.1 THE GEOGRAPHIC EFFECT ON THE GREEN PREMIUM

The mean SHAP values reported in Table 5.7 and Figure 5.9 extend the monetary attribution framework established in Section 5.1 to a regional dimension. For each cell, the figure represents the average isolated contribution of the EPC rating to the predicted price across all transactions in that region-band combination, with all confounding covariates held constant in the Shapley sense. Three findings of substantive interest emerge from this disaggregation.



Table 5.7 - Regional Comparison of Energy Certificate Impact (£)

EPC Band	Cornwall	Greater London	Greater Manchester
G	-7,533.66	-16,454.04	-8,850.54
F	+1,475.35	-1,204.82	-873.43
E	+552.19	+1,554.32	-236.96
D	-478.05	+253.68	-239.30
C	-2,233.03	-1,598.11	-1,704.45
B	+863.00	+3,250.84	+4,288.25
A	+11,871.19	+12,153.14	+16,817.63

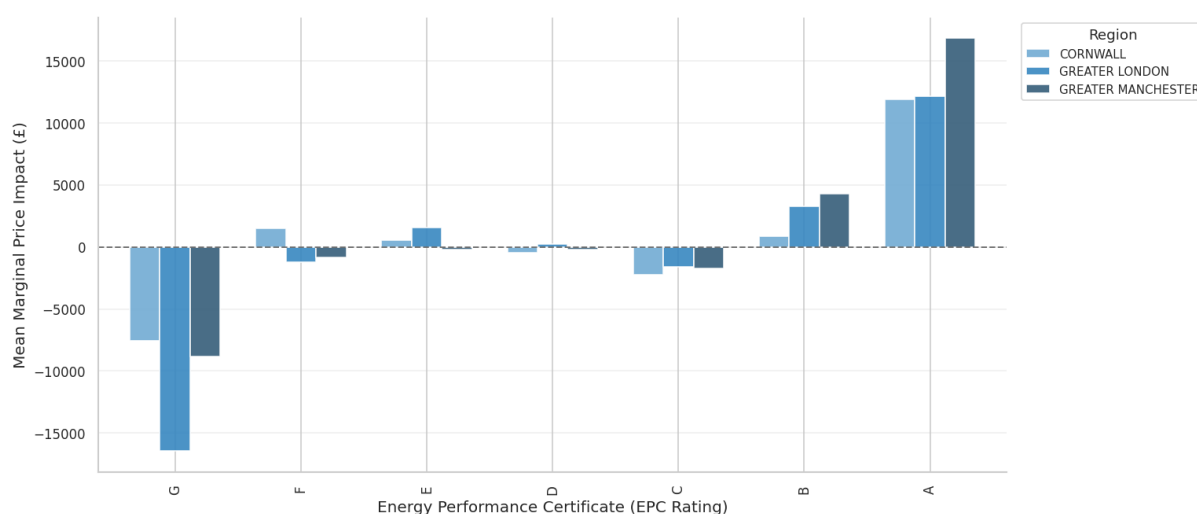


Figure 5.9 - Regional Comparison - EPC Impact on Price by Region (£)

Three findings of substantive interest emerge from this disaggregation.

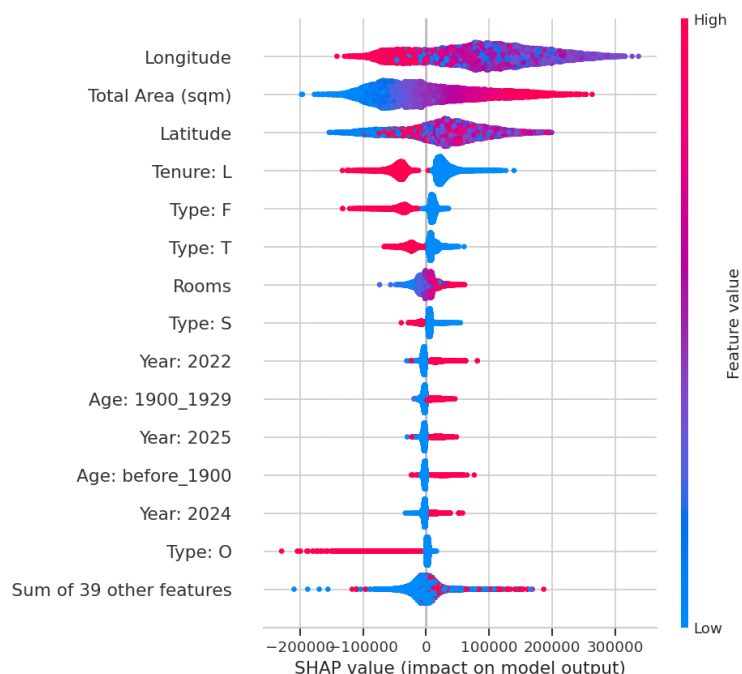
- **The urban efficiency penalty (Band G):** The most severe penalty for low efficiency is observed in Greater London, where Band G properties incur an average SHAP-attributed price reduction of -£16,454. This is approximately double the penalties recorded in Greater Manchester (-£8,851) and Cornwall (-£7,534). The magnitude of the London penalty is consistent with the higher baseline transaction costs associated with energy retrofitting in a high-density urban environment, such as elevated labour costs, restricted access for works, and stricter planning constraints, all of which are capitalised into buyer expectations and therefore reflected in the model's learned attribution.
- **The paradox at the upper efficiency tier (Band A):** Contrary to the pattern observed at the lower end of the scale, the highest absolute premium for Band A properties is found not in London but in Greater Manchester (+£16,818), which marginally exceeds Greater London (+£12,153) and Cornwall (+£11,871). This result is consistent with what may be termed a location price compression effect:



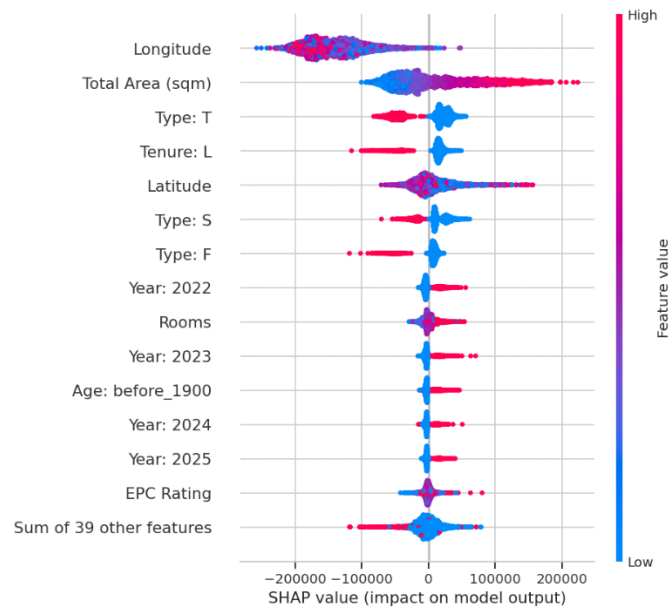
in hyper-valued markets such as Greater London, the dominant explanatory variable is location, which commands so large a share of the model's prediction that the marginal contribution attributable to energy efficiency is proportionally compressed. In mid-priced expanding urban markets, where location constitutes a smaller fraction of total predicted price, the model has greater capacity to attribute price variation to energy performance, producing a higher absolute SHAP contribution for top-rated properties.

- **The non-monotonic intermediate bands (C through E):** The mean SHAP values for Bands C, D, and E do not exhibit a smooth monotonic gradient across regions, and in several cells carry small negative values despite representing above-average efficiency. This is consistent with the well-established finding in hedonic pricing literature that the green premium is concentrated at the extremes of the efficiency distribution, particularly at Band A and Band G, while the intermediate bands are sufficiently prevalent in the housing stock that they carry limited pricing signal relative to the reference category.

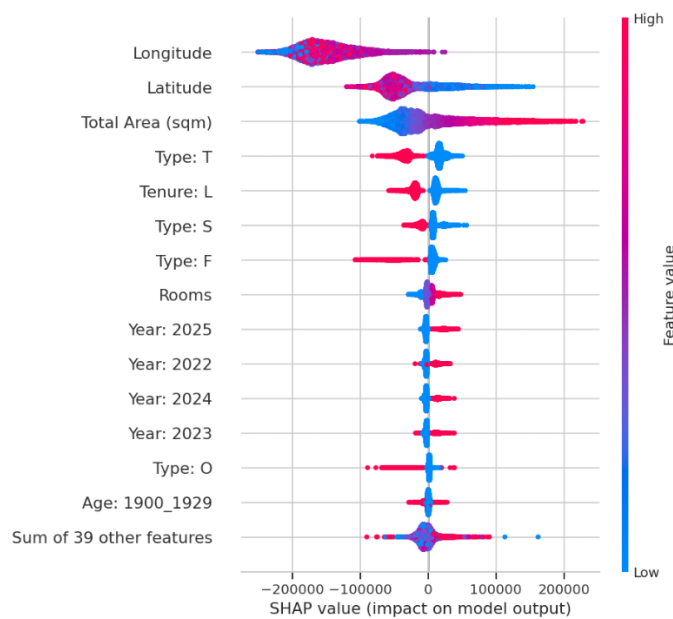
Taken together, these results indicate that the magnitude of the energy efficiency premium is conditioned by the prevailing market context, and that the relationship between EPC rating and price is neither uniform nor strictly monotonic across regions. The SHAP decomposition methodology employed here isolates this effect from confounding covariates with a rigour that conventional subgroup regression would not achieve.



a)



b)



c)

Figure 5.10 - Regional SHAP Summary Plots a) Greater London; b) Cornwall; c) Greater Manchester

- Greater London (The Vertical Metropolis):** The London market is characterised by extreme urban density. Beyond spatial coordinates and floor area, the variables dictating price are fundamentally structural and legal: *Tenure_L* (Leasehold) and *Type_F* (Flats/Maisonettes). The model perfectly captured that London is dominated by high-density apartments transacted under a leasehold regime, making this the main driver of price variance compared to the rest of the country (Figure 5.10 a)).



- **Cornwall (The Premium for Space):** The contrast with the capital is absolute. In Cornwall, the model heavily penalises high-density living and restrictive ownership, pushing *Type_T* (Terraced) and *Tenure_L* (Leasehold) to the top of the hierarchy. By strongly discounting these characteristics, the algorithm mathematically rewards their absence, confirming that Freehold Detached housing dictates the asset's wealth. Notably, this is the only region where the baseline EPC Rating manages to reach the top 14 individual drivers. This describes a rural, coastal market where privacy, the absence of adjacent neighbours, and energy efficiency define the premium (Figure 5.10 b)).
- **Greater Manchester (Suburban Growth):** Manchester exhibits the classic DNA of a large urban agglomeration undergoing rapid demographic expansion. The typologies for terraced (*Type_T*) and semi-detached (*Type_S*) take control of the price structure, pushing detached houses to lower positions. The model statistically translated the sociological reality of English industrial suburbs: a market sustained by contiguous housing tailored for the working and middle classes (Figure 5.10 c)).

5.4 DISCUSSION AND POLICY IMPLICATIONS

The empirical results presented in Chapter 5 collectively advance the understanding of the UK residential property market along three interconnected axes:

1. The methodological superiority of non-linear ensemble learning over the classical Hedonic Pricing Model (HPM);
2. The precise quantification of the environmental margin embedded in EPC ratings;
3. The finding that the financial value of energy efficiency is fundamentally heterogeneous across geographies and architectural eras.

Each of these contributions warrants systematic engagement with the existing academic literature, which this section undertakes in turn.

5.4.1 THE METHODOLOGICAL SHIFT FROM HEDONIC PRICE MODELS TO ENSEMBLE LEARNING

The theoretical foundation upon which this study builds is Rosen's (1974) seminal formulation of hedonic pricing, wherein the market price of a differentiated good, such as a residential property, can be decomposed into the implicit prices of its constituent utility-bearing attributes. Operationalised through OLS regression, the HPM has constituted the canonical instrument of real estate economics for five decades. However, its reliance on linearity, homoscedasticity, and the statistical independence of observations represents assumptions that the UK housing market systematically violates. Fik et al. (2003) formally demonstrated that housing prices exhibit significant



spatial variation incompatible with a fixed-coefficient regression surface, whilst Wheeler and Tiefelsdorf (2005) established that local regression coefficients in geographically heterogeneous markets are subject to multicollinearity that further destabilises OLS estimates. Feng et al. (2021) extended this critique by documenting significant non-linear relationships between house size and price, confirming that the functional forms assumed by conventional HPMs systematically misrepresent the market's true valuation structure.

The OLS baseline deployed in this study, yielding $R^2 = 0.7458$, $MAE = £79,449.17$, and $MAPE = 24.82\%$, is entirely consistent with these theoretical concerns. Whilst an R^2 of approximately 0.75 constitutes an acceptable performance for a parsimonious linear model trained on a national dataset of 514,345 transactions, the magnitude of the residual error, an average absolute deviation of approximately £79,449 per transaction, reflects structural limitations that are architectural rather than data-driven. The champion XGBoost model's improvements are statistically and economically meaningful: $R^2 = 0.8249$, $MAE = £60,868.46$, $RMSE = £94,954.19$, and $MAPE = 17.67\%$. The MAE reduction of approximately £18,581 per transaction (a proportional gain of 23.4%) and the MAPE improvement of 7.15 percentage points confirm that the model's superiority is systematic across the price distribution rather than being concentrated in outlier observations. These results are directly consistent with Li et al. (2021), whose XGBoost-HPM hybrid architecture demonstrated that gradient boosting captures attribute interaction effects, particularly between location, structural type, and temporal conditions, that conventional hedonic specifications cannot represent. In the analogous context of the Hong Kong property market, Deng and Zhang (2025) similarly found that an ensemble learning pipeline augmented by explainable AI outperformed all linear benchmarks in both predictive accuracy and the granularity of economic insight recovered, a conclusion this study replicates on a substantially larger UK national sample. The comparative evidence from Fonseca et al. (2023), who evaluated the Hedonic Model, Random Forest, and Artificial Neural Network in Lisbon's residential market, further corroborates the structural inferiority of OLS when confronted with the complexity of contemporary urban property data.

The comparative performance of the competing ensemble models, LightGBM ($R^2 = 0.8233$, $MAE = £61,545.29$) and Random Forest ($R^2 = 0.8187$, $MAE = £62,201.62$), provides additional methodological insight. All three tree-based models substantially and uniformly outperform OLS, confirming the systematic advantage of ensemble methods in this domain, as catalogued in the survey of Geerts et al. (2023) and the systematic review of Tekouabou et al. (2024). XGBoost's marginal but consistent superiority over both LightGBM and Random Forest is attributable to its regularised learning objective and second-order gradient approximation—design principles identified by Chen and Guestrin (2016) as conferring particular resilience on large, heterogeneous tabular datasets, precisely the characteristics of the Land Registry Price Paid Data merged with EPC certificates. A further architectural contribution is the decision to represent location through continuous geographic coordinates (latitude and



longitude) rather than categorical administrative proxies. Kang et al. (2021) demonstrated that continuous spatial coordinates enable machine learning models to learn the hyper-local price gradients that administrative zone indicators fundamentally cannot capture, a design principle that directly informs the high explanatory power achieved here. This finding is further corroborated by Zhao et al. (2024), whose multi-source data fusion framework confirms that geospatial granularity is among the most impactful data engineering decisions available to the applied practitioner.

5.4.2 THE GREEN PREMIUM AND BROWN DISCOUNT: QUANTIFICATION AND CONTEXTUALISATION

The core empirical contribution of this dissertation is the isolation of the market's environmental margin with a precision hitherto unavailable to the hedonic literature. The application of TreeSHAP, as operationalised by Tchuente (2024) and Tarasov and Dessoulavy-Śliwiński (2025) in comparable AVM contexts, permits the attribution of price effects to individual EPC bands at the property level, eliminating the confounding that afflicts OLS-based estimates when location and energy efficiency are spatially correlated.

The results confirm unambiguously the existence of both a Green Premium and a Brown Discount. Achieving Band A generates an average direct premium of +£13,506.04, and Band B captures +£3,366.37. At the opposite extreme, Band G properties suffer an average penalty of -£12,017.23. The near-zero impact of Band D (+£54.96) is theoretically elegant: the market has internalised the EPC median as its economic baseline, treating Band D as the reference standard against which all premiums and discounts are measured. This result directly extends the empirical work of Chi et al. (2021), whose attribute-linked dataset for England and Wales established the statistical association between EPC ratings and transaction prices but whose OLS-constrained methodology could not isolate the non-linear structure of the premium across the full EPC spectrum that the present SHAP decomposition reveals. Wenninger et al. (2022) demonstrated, using XAI methods on German residential data, that energy performance certificates carry capitalisable value that OLS-based AVMs systematically underestimate, a methodological concern the TreeSHAP approach directly resolves. Oust et al. (2023) similarly found that dwelling condition attributes exert explanatory power that conventional AVM specifications fail to fully capture, echoing this study's finding that the OLS baseline systematically misprices the environmental margin by conflating it with correlated structural attributes.

A theoretically novel contribution is the identification of the "Historical Forgiveness" effect: for properties constructed prior to 1900, the anticipated Brown Discount for low EPC bands (E, F, and G) is substantively attenuated. This pattern implies that buyers assign a heritage premium that partially or fully offsets the energy obsolescence penalty, demonstrating that the market evaluates the cost of retrofitting differently depending on the architectural era of the asset. This interaction between construction age and energy rating is precisely the category of non-linear conditional effect that



XGBoost captures through deep tree branching, and which OLS, with its fixed-coefficient architecture, is constitutionally incapable of detecting. Oust et al. (2023) identified a related mechanism in their analysis of dwelling condition in automated valuation models, whilst Deng and Zhang (2025) explicitly identified the capacity of tree-based ensembles to recover complex interaction effects as a defining advantage of the XGBoost-SHAP methodology, a conclusion that the Historical Forgiveness effect independently validates on UK data.

Crucially, the present data reveal that the Historical Forgiveness mechanism is not merely attenuating in character: at its most extreme, it effects a complete sign reversal of the Brown Discount. Property #31531—a large, freehold, pre-1900 detached house of 138.6 m² in the rural Cornish interior—demonstrates this with mathematical precision: its EPC SHAP contribution is +£41,316.78, representing a deviation of +£53,334 above the national Band G mean of −£12,017. The distribution of the Brown Discount is therefore not merely right-skewed around a negative mean, in a specific combination of rural geography, architectural heritage, and property scale, the penalty reverses sign entirely. The market for such assets is self-selecting, buyers actively seeking pre-industrial stone construction in a remote coastal setting interpret the original unmodified fabric—the very source of the poor thermal rating—as a marker of heritage integrity, not a liability to be priced in. This finding implies that a policy instrument calibrated on the average Band G penalty, or even on the national distributional mean, would fundamentally mischaracterise the economic incentive structure facing owners of this property class.

5.4.3 SPATIAL HETEROGENEITY AND THE LIMITS OF UNIVERSAL EPC POLICY

A principal finding of this study is that the financial value of energy efficiency is neither uniform nor portable across the UK's geographically segmented housing markets. The regional SHAP decomposition reveals systematic cross-market variation that fundamentally challenges the premise of a flat national subsidy for residential energy retrofitting.

Greater London exhibits the most severe Brown Discount nationally, with the average SHAP penalty for Band G properties reaching −£16,454.04, exceeding the national average of −£12,017.23 by approximately 37%. This extreme valuation reflects the capital's aggressive pricing of anticipated future retrofitting costs, consistent with the policy-anticipation mechanism theorised by Wenninger et al. (2022). Paradoxically, the Green Premium for Band A in London (+£12,153.14) is markedly lower than in Greater Manchester (+£16,817.63), despite the capital's substantially higher baseline prices. This inversion—a higher Brown Discount coexisting with a lower Green Premium in the highest-price market—constitutes what this study characterises as the "location price squeeze" effect: when a property's baseline value is anchored to extreme urban land rents, the marginal willingness to pay for environmental certification is constrained by the total budget already committed to location. The gap between London and Manchester is considerable: Greater Manchester's Band A premium exceeds that of



London by approximately £4,664, or 38.4%, despite substantially lower average transaction prices, a compression that becomes economically intelligible only through the lens of the location price squeeze. Kang et al. (2021) documented a structurally analogous dynamic in Chinese high land-value cores, finding that the premium for environmental attributes is elastically compressed in markets where location rents dominate the price structure. This mechanism operates with equal force in the UK context.

The contrast with Greater Manchester is theoretically instructive. Manchester's higher absolute Green Premium for Band A, despite lower baseline prices, reflects a market in which energy efficiency functions as a primary differentiation mechanism. In suburban and post-industrial growth markets, the pricing of environmental quality is both more elastic and less distorted by the primacy of land value. This regional divergence is consistent with the findings of Karamanou et al. (2022), who demonstrated, using SHAP-explained open government data applied to Scottish housing markets, that the determinants of price heterogeneity are fundamentally regional in character and cannot be reliably approximated by national aggregates. The present study extends this insight specifically to the pricing of EPC ratings, establishing empirically that willingness to pay for energy efficiency varies not merely quantitatively but qualitatively across the UK urban hierarchy.

From a policy perspective, the evidence assembled in this study carries a clear imperative: blanket national subsidies for energy retrofitting will be geographically misallocated by design. In high-cost urban markets such as Greater London, where affordability constraints constitute the binding limitation on sustainable investment, uniform national incentives will provide the least leverage precisely where the Brown Discount is most severe. Furthermore, the extreme outlier cases identified through individual-level SHAP analysis, such as a single Band G Victorian terraced property in Greater London attracting a SHAP penalty approaching -£81,000, demonstrate that the absolute magnitude of the Brown Discount is highly variable and context-dependent. Simultaneously, the Cornwall Heritage outlier (+£41,317 for a Band G property) establishes that the distributional tails extend in both directions: the Brown Discount is not merely variable in magnitude, but variable in sign. Aggregate averages, therefore, conceal both pockets of concentrated obsolescence risk and pockets of heritage premium that demand opposing policy responses, the former requiring targeted remediation support, the latter requiring explicit carve-outs to avoid disincentivising the preservation of unmodified period architecture. Future urban governance frameworks must therefore adopt regionally calibrated financial instruments, a design principle articulated in the broader agenda for explainable AI in urban decision-making by Trindade Neves et al. (2024) and Tarasov and Dessoulavy-Śliwiński (2025). The XGBoost-SHAP pipeline developed in this study offers precisely the property-level, geographically granular evidence base required for such targeted policy calibration, translating the abstract imperative of the UK's Net Zero 2050 strategy into actionable, market-consistent financial interventions.



6. CONCLUSIONS AND FUTURE RESEARCH

This dissertation set out to decode the complex dynamics of the UK housing market between 2021 and 2025, moving beyond traditional econometric limitations by intersecting OGD, advanced ML, and XAI. By systematically addressing the core research objectives defined at the onset of this study, the following empirical conclusions were drawn:

The first objective sought to evaluate whether tree-based ensemble algorithms could outperform the classical HPM (OLS) in predicting property values. The empirical results unequivocally validate this hypothesis. The real estate market is inherently non-linear and hyper-local, a complexity that linear models fail to capture. The eXtreme Gradient Boosting algorithm (XGBoost), optimised via Bayesian Optimisation, emerged as the definitive predictive champion. It achieved an R^2 of 0.8249 and reduced the Mean Absolute Error (MAE) to £60,868, effectively decreasing the average prediction error by approximately £18,581 per transaction when compared to the OLS baseline. This proves that advanced algorithms, particularly those leveraging exact spatial coordinates (latitude and longitude) rather than broad administrative zones, are essential for modern AVMs.

The central research question aimed to overcome the "black-box" nature of ML to isolate the exact financial value of EPC. Through the application of the SHAP framework, the model successfully quantified the market's environmental margin. The analysis confirms the undeniable existence of both a "Green Premium" and a "Brown Discount". Reaching the highest efficiency tier (Band A) yields an average direct premium of +£13,506 to a property's transaction value, whilst Band B secures a +£3,366 premium. Conversely, extreme energy obsolescence (Band G) is heavily penalised, imposing an average discount of -£12,017. Furthermore, Band D demonstrated a marginal impact of virtually zero (+£55), corroborating the economic theory that the market has fully internalised this tier as the absolute baseline standard.

The final objective was to determine if the valuation of energy efficiency is uniform across different geographies and architectural eras. The local interpretability extracted via SHAP firmly rejects the notion of a universal, flat-rate premium. The valuation of sustainability is highly elastic and context-dependent: a) Spatial Squeeze: In highly tensioned urban markets like Greater London, the fundamental cost of location depletes the buyers' budget, constraining the maximum absolute "Green Premium" whilst concurrently imposing the nation's most severe "Brown Discount" (exceeding -£16,400 for Band G properties). Conversely, expanding suburban markets like Greater Manchester exhibited the highest absolute willingness to pay for Band A efficiency (+£16,818); and b) Historical Forgiveness: The intersectional analysis revealed a sophisticated, bifurcated market behaviour regarding historical housing stock (built before 1900). In some instances, the market actively "forgives" energy obsolescence in exchange for Victorian or Edwardian architectural charm. However, for massive properties in central hubs, this same era triggers severe penalties (exceeding -£80,000



in isolated cases), as the market fully capitalises in the immense capital expenditure and regulatory complexity required for modern retrofitting.

In summary, this research successfully bridges the gap between high-accuracy prediction and deep economic interpretability, providing a comprehensive, data-driven anatomy of how sustainability is priced into the modern UK real estate ecosystem.

This dissertation delivers substantial contributions to both the academic literature of real estate economics and the practical application of AVMs in the proptech industry.

From a theoretical standpoint, this study challenges the traditional reliance on HPMS. By proving that the market valuation of energy efficiency is highly non-linear and subject to extreme spatial heterogeneity, it establishes that linear methodologies (OLS) consistently misprice the "Green Premium". Furthermore, the discovery of the bifurcated reality for 19th-century architecture adds a new behavioural dimension to environmental economics, demonstrating that cultural heritage can either temporarily override climate-driven depreciation or severely amplify the "retrofitting debt."

For industry practitioners (banks, real estate funds, and valuation agencies), the developed XGBoost-SHAP pipeline offers a highly accurate, deployable AVM capable of assessing property-specific retrofitting risks down to the exact geographical coordinates. For policymakers and urban planners, the evidence of a "location price squeeze" (where extreme baseline prices in London compress the available budget for green premiums) serves as a critical warning, indicating that blanket national subsidies for energy retrofitting will be inefficient. Instead, financial incentives must be regionally calibrated to address the specific market tensions of each geographical area.

While the deployed methodology achieved high predictive accuracy (R^2 of 0.8249) using strictly open-source data, certain limitations inherent to the dataset and scope must be acknowledged. Firstly, the dataset lacks qualitative variables regarding the interior condition of the properties. The model assumes that unexplained variance is partially attributed to these unobserved characteristics, which can significantly sway individual transaction prices. Secondly, although the model captures temporal inflation via *Year_of_Transfer* and *Month_of_Transfer*, it does not explicitly integrate high-frequency macroeconomic indicators such as daily mortgage interest rates, specific inflation indices, or shifting government tax policies, which actively influenced buyer purchasing power between 2021 and 2025. Thirdly, the LR-PPD dataset only records successful property transactions. Therefore, the model cannot capture the dynamics of the rental market or, crucially, properties that suffered from "green obsolescence" and failed to sell due to poor energy ratings.

The findings and limitations of this study pave the way for several highly promising avenues for future research. Future iterations could significantly enrich the dataset by merging OGD with Geographic Information Systems (GIS). Calculating exact distances to key points of interest, such as underground stations, top-rated schools, or green spaces, could squeeze further residual error out of the XGBoost model. While Gradient



Boosting is currently the gold standard for tabular data, future studies could explore novel Deep Learning architectures designed specifically for tabular structures (such as TabNet) to evaluate if neural networks can surpass tree-based ensembles in capturing spatial heterogeneity. As the UK government enforces stricter EPC regulations approaching the 2050 Net Zero target, future research could apply dynamic time-series forecasting to predict how the "Brown Discount" will evolve. It is hypothesised that the penalty for Band F and G properties will increase exponentially as legal deadlines for mandatory retrofitting draw closer, a trend that warrants continuous algorithmic monitoring.



BIBLIOGRAPHICAL REFERENCES

- Allam, Z., & Dhunny, Z. A. (2019). On big data, artificial intelligence and smart cities. *Cities*, 89, 80–91. <https://doi.org/10.1016/j.cities.2019.01.032>
- Bibri, S. E., Alexandre, A., Sharifi, A., & Krogstie, J. (2023). Environmentally sustainable smart cities and their converging AI, IoT, and big data technologies and solutions: An integrated approach to an extensive literature review. *Energy Informatics*, 6(1), 9. <https://doi.org/10.1186/s42162-023-00259-2>
- Cellmer, R. (2023). Points of Interest and Housing Prices. *Real Estate Management and Valuation*, 31(1), 69–77. <https://doi.org/10.2478/remav-2023-0007>
- Chen, N. (2022). House Price Prediction Model of Zhaoqing City Based on Correlation Analysis and Multiple Linear Regression Analysis. *Wireless Communications and Mobile Computing*, 2022(1), 9590704. <https://doi.org/10.1155/2022/9590704>
- Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, KDD '16*, 785–794. <https://doi.org/10.1145/2939672.2939785>
- Chi, B., Dennett, A., Oléron-Evans, T., & Morphet, R. (2021). A new attribute-linked residential property price dataset for England and Wales, 2011–2019. *UCL Open Environment*, 3(1). <https://doi.org/10.14324/111.444/ucloe.000019>
- Deng, L., & Zhang, X. (2025). Boosting the accuracy of property valuation with ensemble learning and explainable artificial intelligence: The case of Hong Kong. *The Annals of Regional Science*, 74(1), 32. <https://doi.org/10.1007/s00168-025-01365-7>
- Feng, S.-T., Peng, C.-W., Yang, C.-H., & Chen, P.-W. (2021). Non-linear relationships between house size and price. *International Journal of Strategic Property Management*, 25(3), 240–253. <https://doi.org/10.3846/ijspm.2021.14607>
- Fik, T. J., Ling, D. C., & Mulligan, G. F. (2003). Modeling Spatial Variation in Housing Prices: A Variable Interaction Approach. *Real Estate Economics*, 31(4), 623–646. <https://doi.org/10.1046/j.1080-8620.2003.00079.x>
- Fonseca, P., Jardim, B., & de Castro Neto, M. (2023). *Hedonic Model, Random Forest and Artificial Neural Network: Comparison for Real Estate Price Prediction in Lisbon* (SSRN Scholarly Paper No. 4363508). Social Science Research Network. <https://doi.org/10.2139/ssrn.4363508>
- Geerts, M., vanden Broucke, S., & De Weerd, J. (2023). A Survey of Methods and Input Data Types for House Price Prediction. *ISPRS International Journal of Geo-Information*, 12(5), 200. <https://doi.org/10.3390/ijgi12050200>



- Hong, J., & Kim, W.-S. (2022). Combination of machine learning-based automatic valuation models for residential properties in South Korea. *International Journal of Strategic Property Management*, 26, 362–384. <https://doi.org/10.3846/ijspm.2022.17909>
- Kalampokis, E., Zeginis, D., & Tarabanis, K. (2018). *On Modeling Linked Open Statistical Data* (SSRN Scholarly Paper No. 3287147). Social Science Research Network. <https://doi.org/10.2139/ssrn.3287147>
- Kang, Y., Zhang, F., Peng, W., Gao, S., Rao, J., Duarte, F., & Ratti, C. (2021). Understanding house price appreciation using multi-source big geo-data and machine learning. *Land Use Policy*, 111, 104919. <https://doi.org/10.1016/j.landusepol.2020.104919>
- Karamanou, A., Kalampokis, E., & Tarabanis, K. (2022). Linked Open Government Data to Predict and Explain House Prices: The Case of Scottish Statistics Portal. *Big Data Research*, 30, 100355. <https://doi.org/10.1016/j.bdr.2022.100355>
- Li, S., Jiang, Y., Ke, S., Nie, K., & Wu, C. (2021). Understanding the effects of influential factors on housing prices by combining extreme gradient boosting and a hedonic price model (Xgboost-hpm). *Land*, 10(5). <https://doi.org/10.3390/land10050533>
- Lundberg, S. M., Erion, G., Chen, H., DeGrave, A., Prutkin, J. M., Nair, B., Katz, R., Himmelfarb, J., Bansal, N., & Lee, S.-I. (2020). From Local Explanations to Global Understanding with Explainable AI for Trees. *Nature Machine Intelligence*, 2(1), 56–67. <https://doi.org/10.1038/s42256-019-0138-9>
- Neves, F. T., de Castro Neto, M., & Aparicio, M. (2020). The impacts of open data initiatives on smart cities: A framework for evaluation and monitoring. *Cities*, 106, 102860. <https://doi.org/10.1016/j.cities.2020.102860>
- Oust, A., Westgaard, S., Waage, J. E., & Yemane, N. K. (2023). Assessing the explanatory power of dwelling condition in automated valuation models. *Journal of Real Estate Research*. <https://ntnuopen.ntnu.no/ntnu-xmlui/handle/11250/3105078>
- Rosen, S. (1974). Hedonic Prices and Implicit Markets: Product Differentiation in Pure Competition. *Journal of Political Economy*, 82(1), 34–55.
- Taecharungroj, V. (2021). Google Maps amenities and condominium prices: Investigating the effects and relationships using machine learning. *Habitat International*, 118, 102463. <https://doi.org/10.1016/j.habitatint.2021.102463>
- Tarasov, S., & Dessoulavy-Śliwiński, B. (2025). Algorithm-Driven Hedonic Real Estate Pricing – An Explainable AI Approach. *Real Estate Management and Valuation*, 33(1), 22–34. <https://doi.org/10.2478/remav-2025-0003>



Tchuente, D. (2024). Real Estate Automated Valuation Model with Explainable Artificial Intelligence Based on Shapley Values. *The Journal of Real Estate Finance and Economics*. <https://doi.org/10.1007/s11146-024-09998-9>

Tekouabou, S. C. K., Gherghina, Ș. C., Kameni, E. D., Filali, Y., & Idrissi Gartoumi, K. (2024). AI-Based on Machine Learning Methods for Urban Real Estate Prediction: A Systematic Survey. *Archives of Computational Methods in Engineering*, 31(2), 1079–1095. <https://doi.org/10.1007/s11831-023-10010-5>

Trindade Neves, F., Aparicio, M., & De Castro Neto, M. (2024). The Impacts of Open Data and eXplainable AI on Real Estate Price Predictions in Smart Cities. *Applied Sciences*, 14(5), 2209. <https://doi.org/10.3390/app14052209>

Wenninger, S., Karnebogen, P., Lehmann, S., Menzinger, T., & Reckstadt, M. (2022). Evidence for residential building retrofitting practices using explainable AI and socio-demographic data. *Energy Reports*, 8, 13514–13528. <https://doi.org/10.1016/j.egyr.2022.10.060>

Wheeler, D., & Tiefelsdorf, M. (2005). Multicollinearity and correlation among local regression coefficients in geographically weighted regression. *Journal of Geographical Systems*, 7(2), 161–187. <https://doi.org/10.1007/s10109-005-0155-6>

Yigitcanlar, T., Corchado, J. M., Mehmood, R., Li, R. Y. M., Mossberger, K., & Desouza, K. (2021). Responsible Urban Innovation with Local Government Artificial Intelligence (AI): A Conceptual Framework and Research Agenda. *Journal of Open Innovation: Technology, Market, and Complexity*, 7(1), 71. <https://doi.org/10.3390/joitmc7010071>

Zhao, Y., Zhao, J., & Lam, E. Y. (2024). House Price Prediction: A Multi-Source Data Fusion Perspective. *Big Data Mining and Analytics*, 7(3), 603–620. <https://doi.org/10.26599/BDMA.2024.9020019>



APPENDIX A - SUPPLEMENTARY EXPLORATORY DATA ANALYSIS

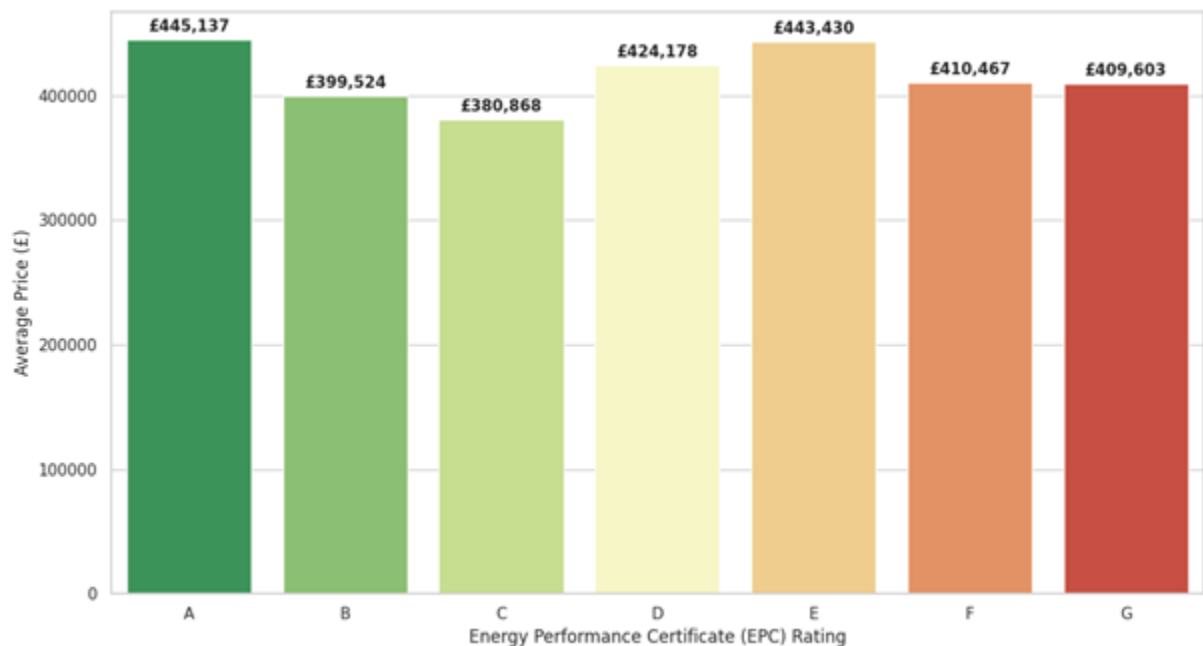


Figure A.1 - Average property prices across Energy Performance Certificate (EPC) ratings. The non-linear trend highlights the premium on both highly efficient new builds (Bands A) and inherently inefficient historical properties (Bands E-G).

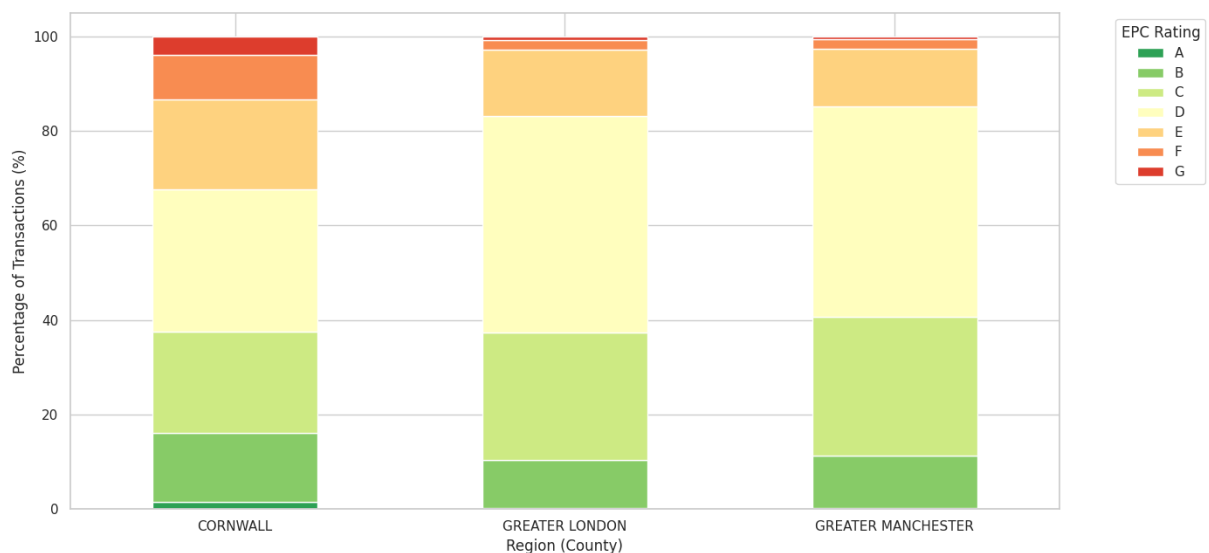


Figure A.2 - Regional distribution of Energy Performance Certificate (EPC) ratings, highlighting the structural disparities between Cornwall, Greater London, and Greater Manchester.

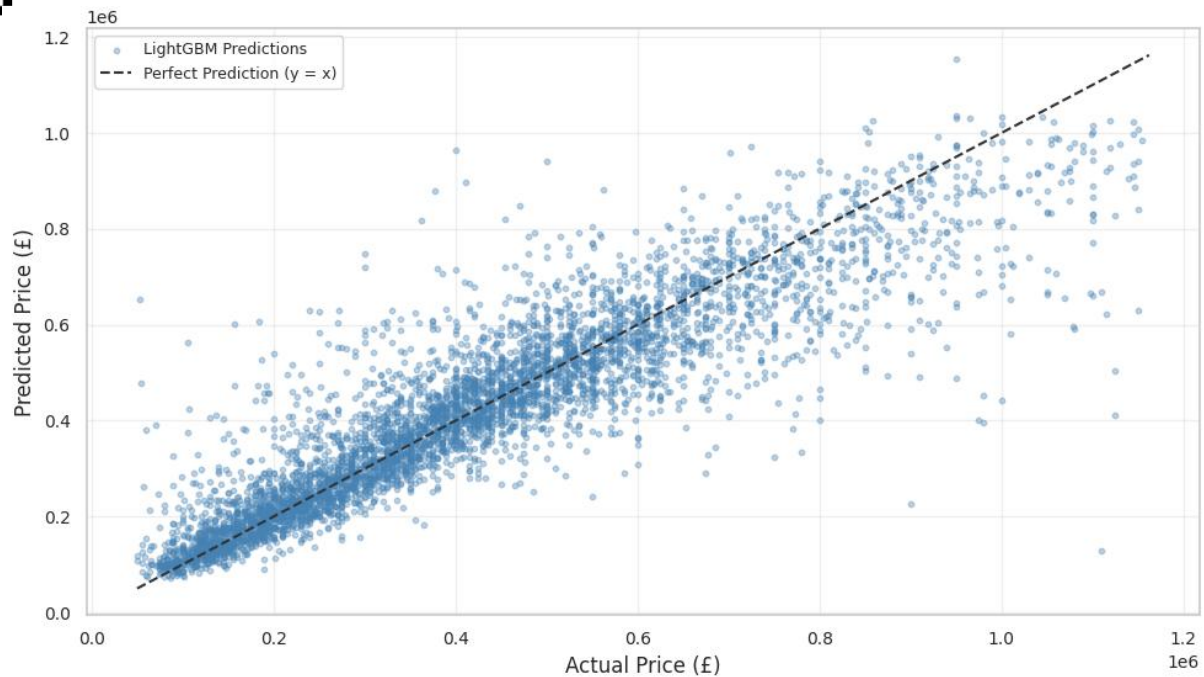


Figure A.3 - Predictive Performance Diagnostic Plot for the optimized LightGBM model: Actual versus Predicted Property Prices (£). The visualisation employs hexagonal binning to illustrate the density of observations, demonstrating a strong linear alignment along the identity line across the test set.

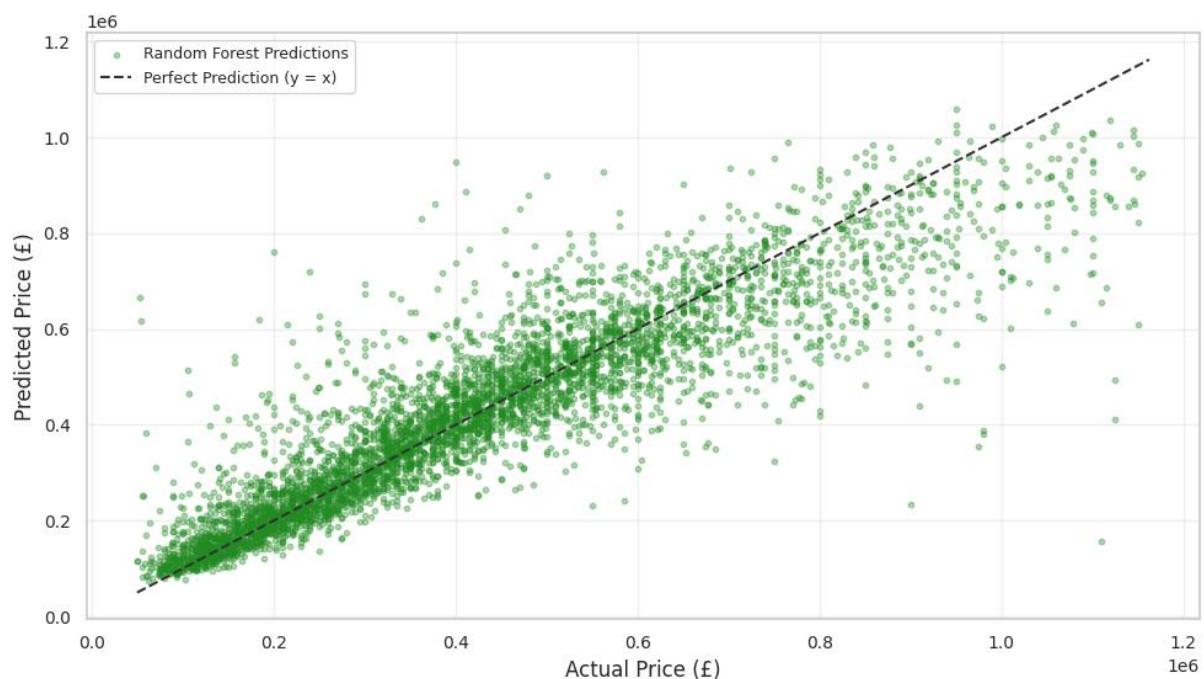


Figure A.4 - Predictive Performance Diagnostic Plot for the optimized Random Forest model: Actual versus Predicted Property Prices (£). Utilising hexagonal binning to represent transaction density, the plot reflects the model's robust predictive accuracy, achieving an R^2 of 0.8187 and a Mean Absolute Error of £62,201 on the test set.

Data with Purpose.

